Gene expression group

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Machine Learning on gene expression data

- Introduction to supervised and unsupervised machine learning concepts.
- Use of a real-world pan-cancer omics dataset from The Cancer Genome Atlas (TCGA).
- Analysis of high-dimensional gene expression data.
- Integration of gene expression data with clinical annotations.
- Dimensionality reduction using methods such as PCA and UMAP.
- Unsupervised learning and clustering (hierarchical clustering, k-means).
- Development of supervised machine learning models for classification tasks.
- Introduction to deep learning approaches for high-dimensional omics data.
- Biological interpretation of machine learning results.

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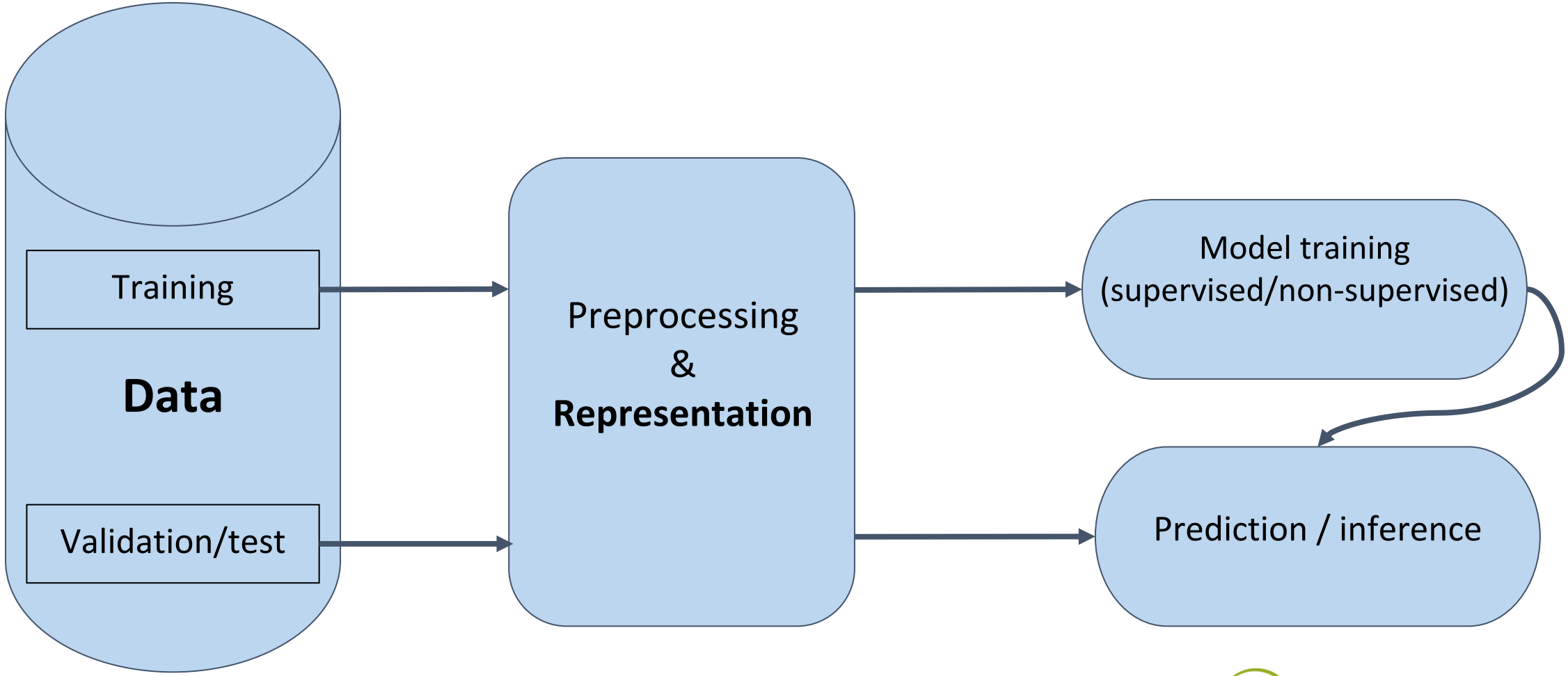
Machine Learning on gene expression data

In this session:

- Dimensionality reduction
- Clustering

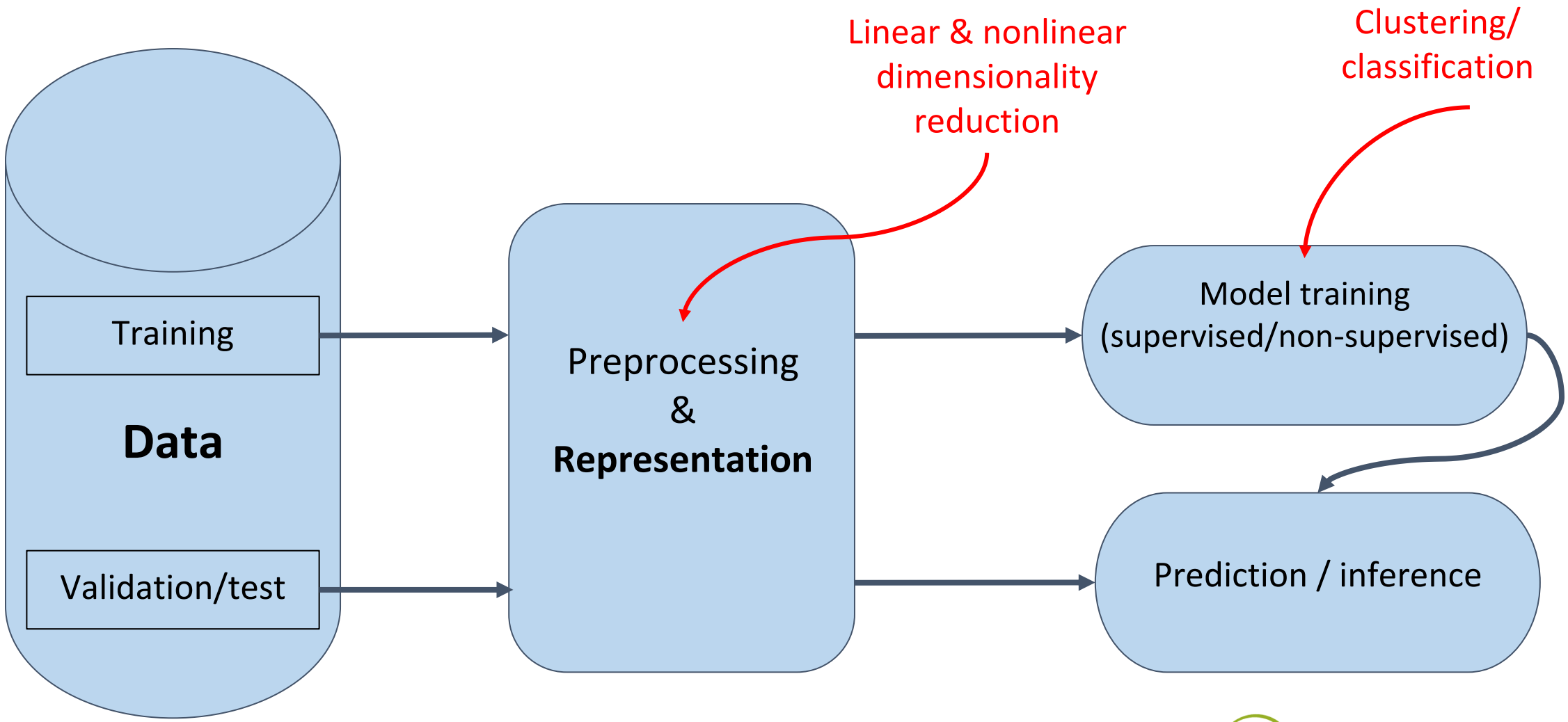
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Machine Learning



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Machine Learning



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What is dimensionality reduction?

Dimensionality reduction is the process of **transforming high-dimensional data** into a **lower-dimensional representation** while **preserving**, as much as possible, the **information** that is **relevant for a given task** (such as prediction, clustering, visualization, or interpretation).

Friedman, J., Hastie, T., & Tibshirani, R. (2009). van der Maaten, L., Postma, E., & van den Herik, J. (2009). Jolliffe, I. T., & Cadima, J. (2016).

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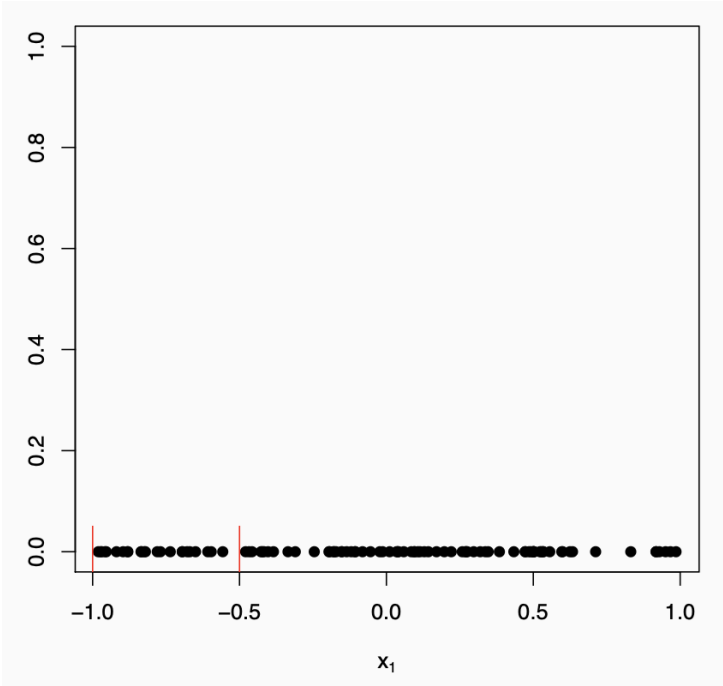
Dimensionality reduction

- Improves generalization and computational efficiency
- Facilitates visualization and interpretability
- Mitigates the curse of dimensionality
- Acts as feature extraction rather than feature selection

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Dimensionality reduction

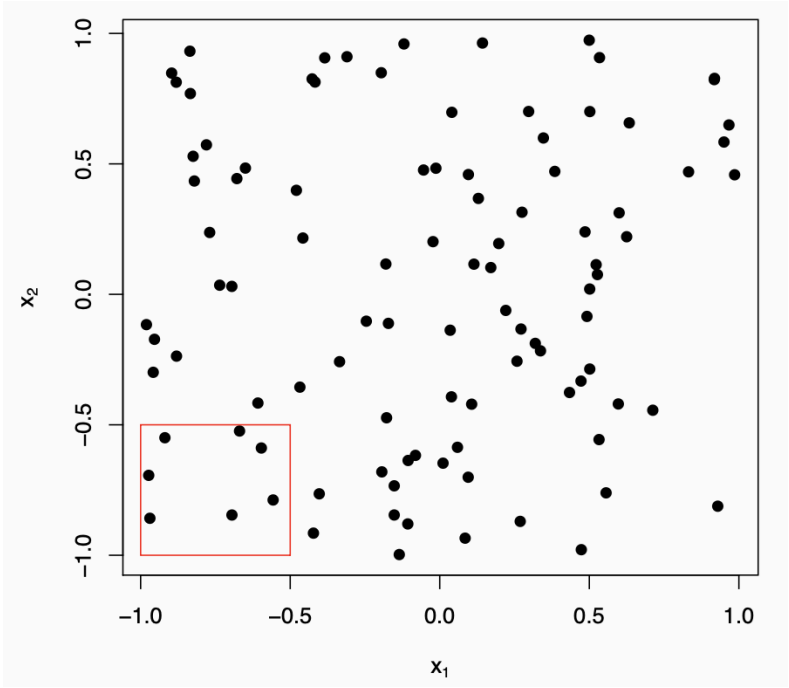
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Dimensionality reduction

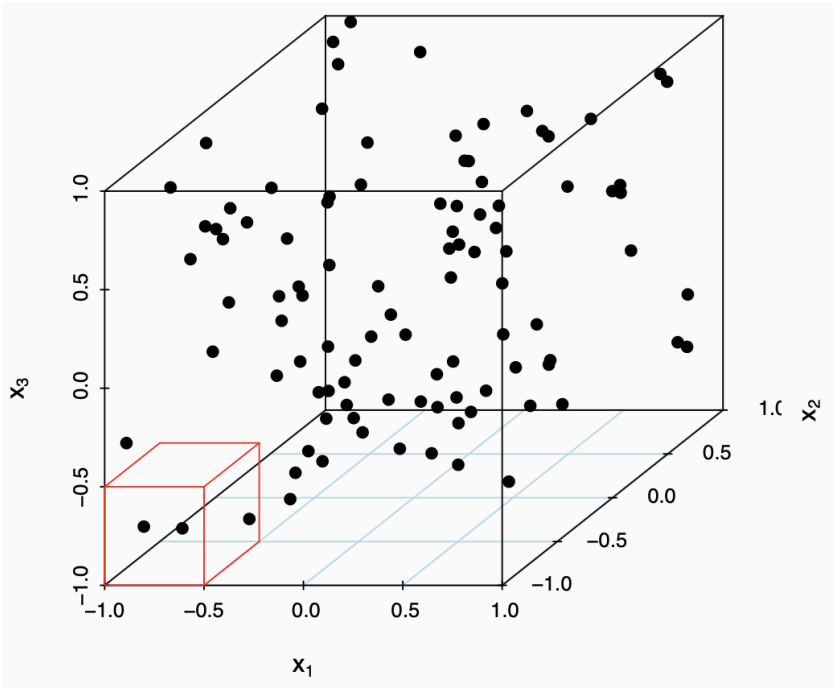
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Dimensionality reduction

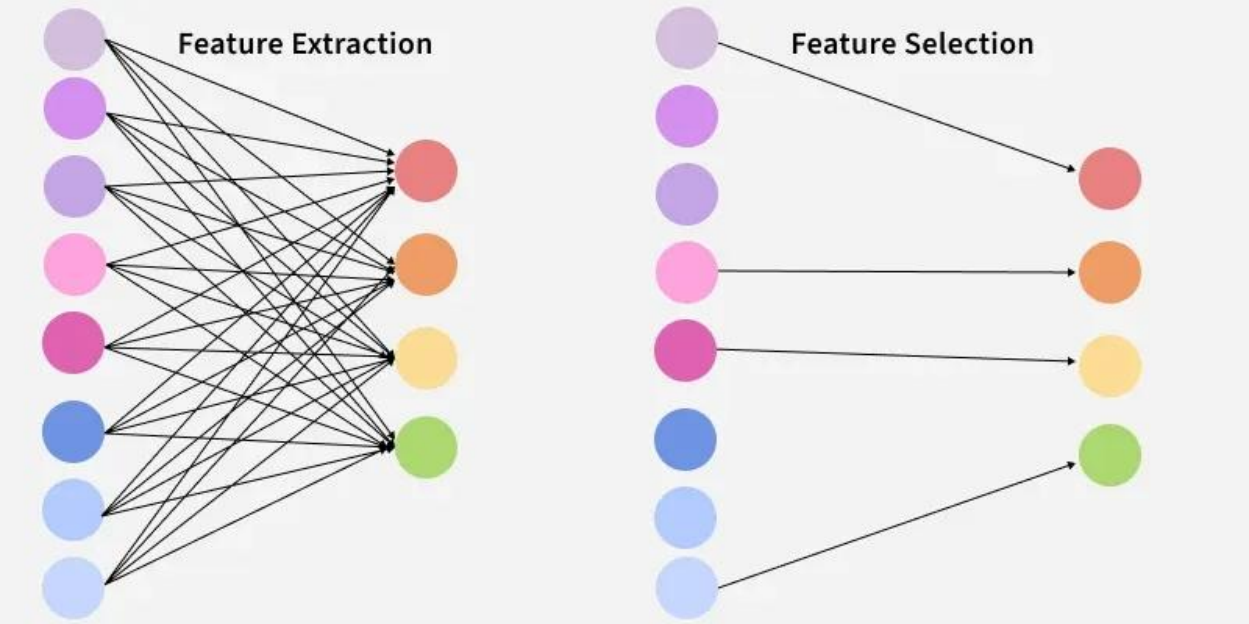
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Dimensionality reduction

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- Mitigates the curse of dimensionality
- **Acts as feature extraction rather than feature selection**



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Dimensionality reduction by feature selection

Filter

Use statistical measures to rank and select features.

- **Correlation Coefficient**
- **Chi-Square Test**
- **ANOVA (Analysis of Variance)**
- **Information Gain**

Wrapper

Use machine learning models to evaluate feature subsets.

- **Forward Selection**
- **Backward Elimination**
- **Bidirectional Elimination**

These methods are accurate but computationally expensive.

Embedded

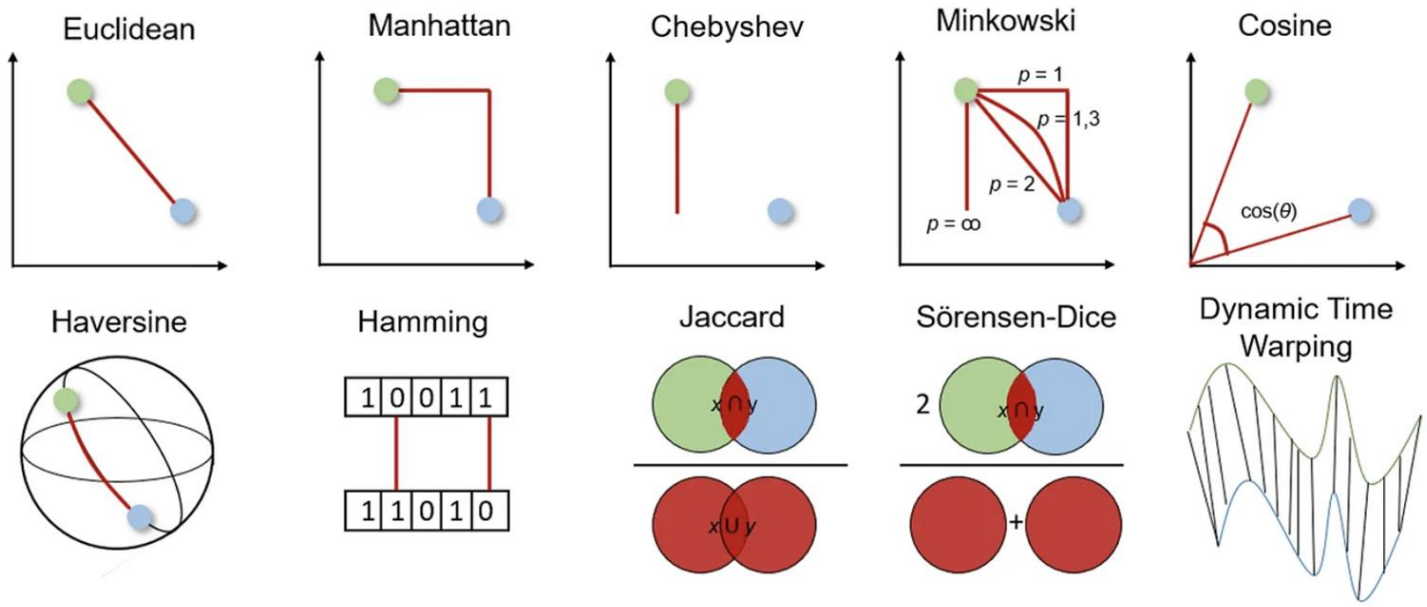
Perform feature selection during model training.

- **LASSO (Least Absolute Shrinkage and Selection Operator)**
- **Ridge Regression**
- **Elastic Net**

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Distances

A distance is a measure of how far apart (or different) two points are. There are different distance functions, but in this module, we will explore the popular ones.



A distance matrix is a simetric, positive-definite matrix were distances between pairs of points are represented according to some distance function.

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Dimensionality reduction

Linear

- Principal Component Analysis (PCA)
- Multidimensional Scaling (Classical MDS)

Nonlinear (manifold learning)
Isomap
Locally Linear Embedding (LLE)
t-Distributed Stochastic Neighbor Embedding (t-SNE)
Uniform Manifold Approximation and Projection (UMAP)

Representation learning
Autoencoders (sparse, denoising, variational, deep)

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Dimensionality reduction by feature extraction

Linear Methods

Principal components (PCA)

capture the maximum variance in the data. used for reducing dimensions while preserving as much variance as possible.

Linear Discriminant Analysis (LDA)

useful for classification tasks, as it reduces dimensions while preserving class separability.

No linear methods (Manifold learning)

t - SNE

for visualizing high-dimensional data in 2D or 3D by preserving local 2D or 3D relationships between data pointsnships between data points

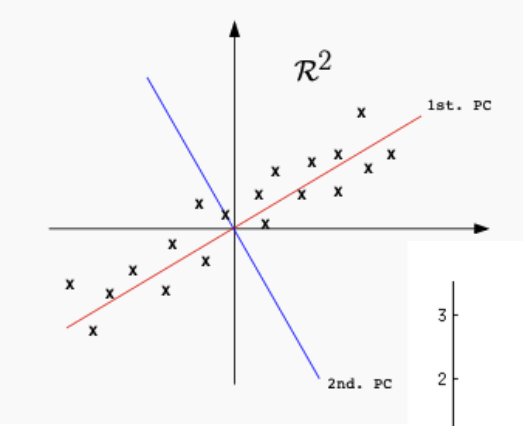
UMAP

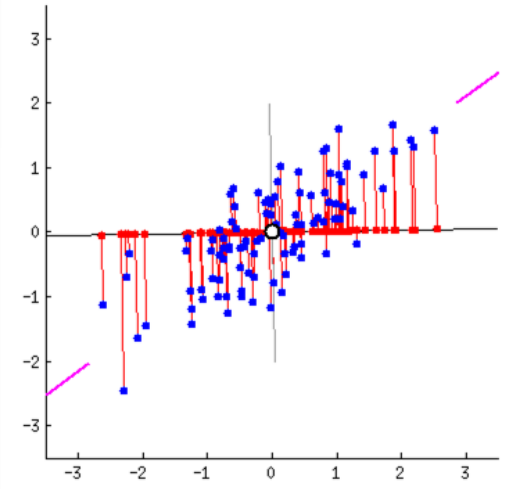
It is similar to t-SNE but is faster and better at preserving global structure.

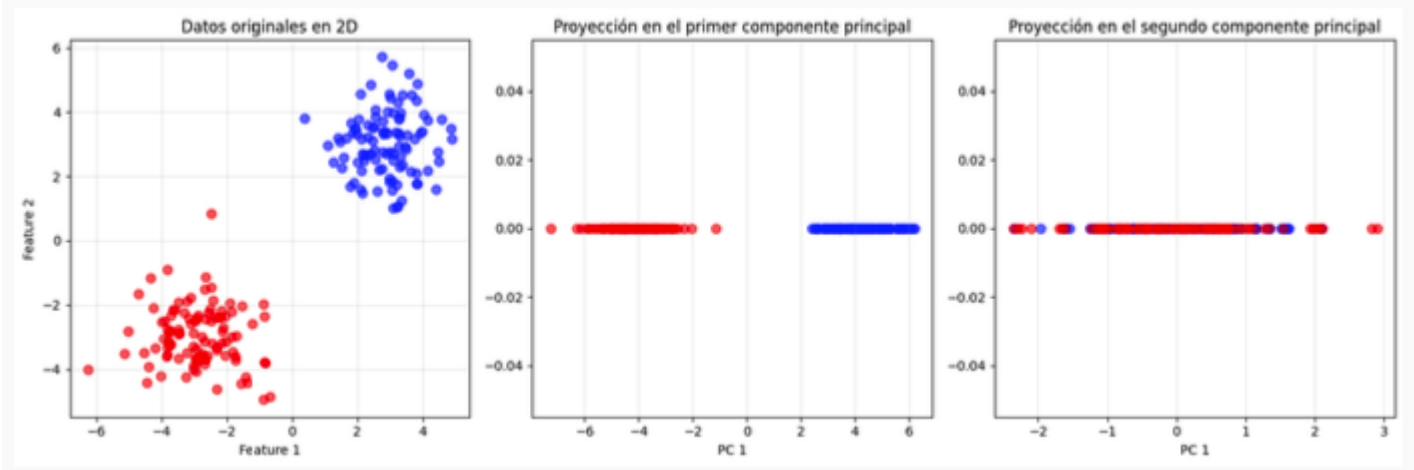
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Principal Component Analysis (PCA)

Idea: Linear dimensionality reduction by projecting data onto orthogonal directions of maximum variance.







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Principal Component Analysis (PCA)

Formulation: given a data matrix with *n* observations and *d* variables (atributes), $\mathbf{X}_{(n \times d)}$ and corresponding covariance matrix

$$\mathbf{S} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X}$$

find projection components **u** such that

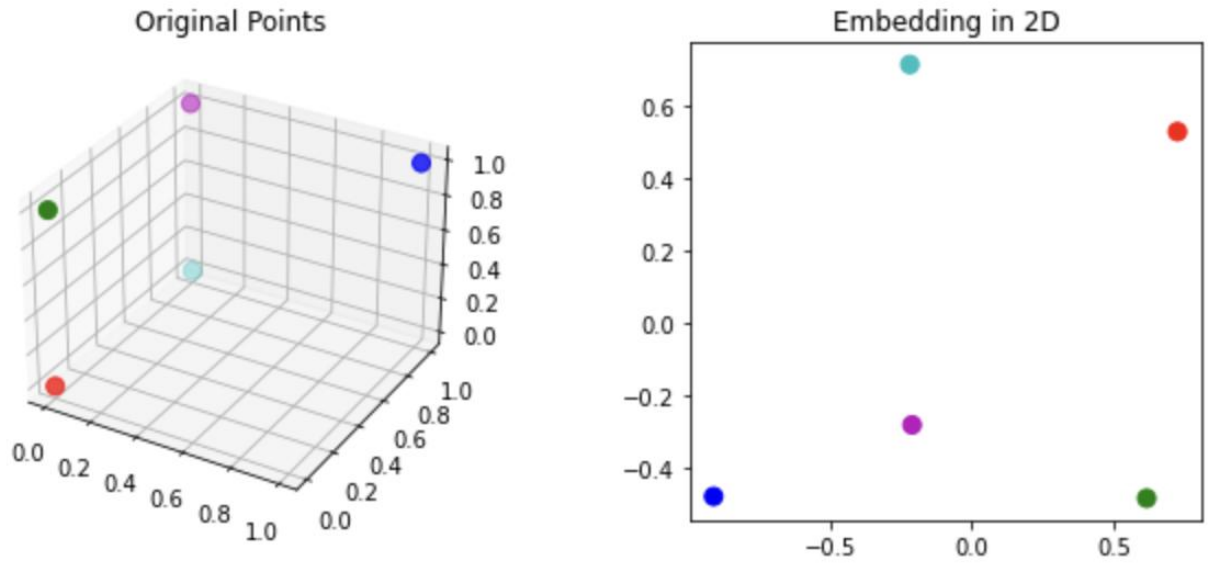
$$\max_{\mathbf{u}} \mathrm{Var}(\mathbf{u}^T\mathbf{X}) = \mathbf{u}^T\mathbf{S}\mathbf{u} \quad \text{sujeto a } \|\mathbf{u}\|^2 = 1,$$

Principal components are eigenvectors of **S** with largest eigenvalues $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$, and project data on those components: $\mathbf{y}_i = \mathbf{u}_i^T\mathbf{X}$

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Multidimensional scaling (MDS) or Principal Coordinate Analysis (PCoA)

Idea: Represents data in a low-dimensional space while preserving pairwise distances (or similarities).



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Classical multidimensional scaling (MDS) or Principal Coordinate Analysis (PCoA)

From X, compute a Distance or Proximity matrix D_{nxn} . MDS finds coordinates *x* that minimizes the Stress function:

$$Stress_D(x_1, x_2, \ldots, x_N) = \sqrt{\sum_{i \neq j = 1,\ldots,N} (d_{ij} - ||x_i - x_j||)^2}$$

The goal of the algorithm is to minimize the value of stress.

Where $x_1, \ldots, x_N$ are data points with their new set of coordinates in lower dimensional space.

d_{ij} is the actual distance we have calculated between the two corresponding data points in their original dimensional space.

$||x_i - x_j||$ is the distance between the two corresponding data points in their lower dimensional space.

The closer the value of $||x_i - x_j||$ is to d_{ij} the smaller will be the value of stress.

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SNE and t-SNE nonlinear dimensionality reduction

Idea:

- Stochastic Neighbor Embedding (SNE) is a nonlinear dimensionality reduction method that preserves local neighborhoods by minimizing the Kullback–Leibler divergence between high-dimensional and low-dimensional similarity distributions.
- t-SNE extends and improves SNE by using symmetric probabilities and a heavy-tailed Student’s t-distribution in the embedding space, yielding clearer separation of local clusters.

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SNE and t-SNE nonlinear dimensionality reduction

- Similarity of data points in High Dimension

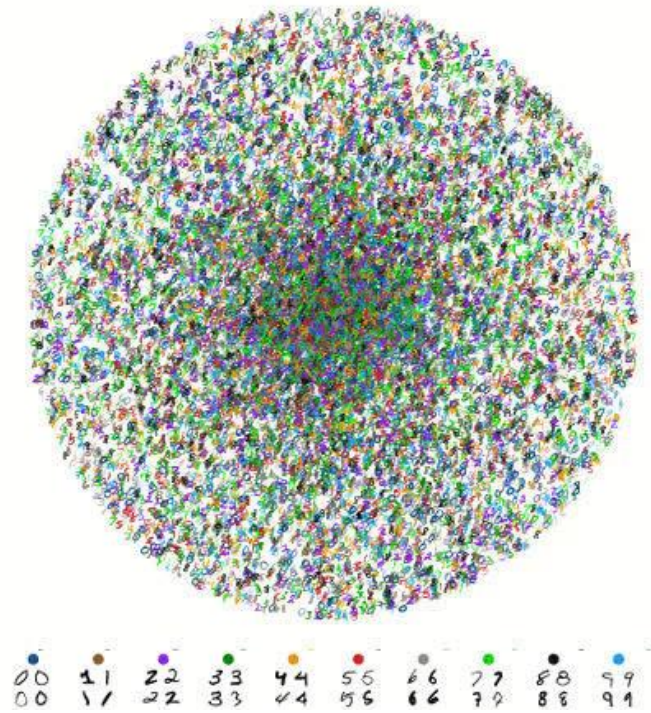
$$p_{j|i} = \frac{\exp\left(-\|x_i - x_j\|^2 / 2\sigma_i^2\right)}{\sum_{k \neq i} \exp\left(-\|x_i - x_k\|^2 / 2\sigma_i^2\right)},$$

- Similarity of data points in Low Dimension

$$q_{j|i} = \frac{\exp\left(-\|y_i - y_j\|^2\right)}{\sum_{k \neq i} \exp\left(-\|y_i - y_k\|^2\right)}.$$

- Cost function

$$C = \sum_i KL(P_i||Q_i) = \sum_i \sum_j p_{j|i} \log \frac{p_{j|i}}{q_{j|i}},$$



https://nicola17.github.io/tfjs-tsne-demo/

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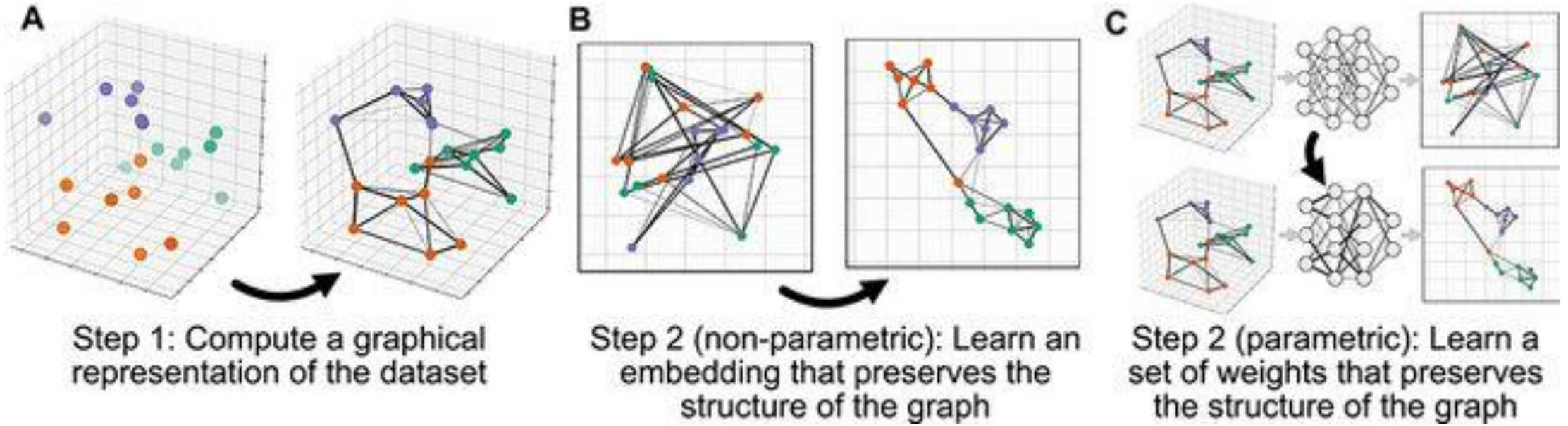
Uniform manifold Uniform Manifold Approximation and Projection (UMAP)

Idea:

- UMAP is a nonlinear dimensionality reduction method that constructs a low-dimensional embedding by modeling local neighborhood relationships as a fuzzy topological structure and optimizing an objective that preserves both local and some global manifold structure, primarily for visualization and representation learning.

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Uniform manifold Uniform Manifold Approximation and Projection (UMAP)



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Advantages and disadvantages to use dimensionality reduction

Advantages of Dimensionality Reduction

- **Faster Computation**: With fewer features machine learning algorithms can process data more quickly. This results in faster model training and testing which is particularly useful when working with large datasets.
- **Better Visualization**: As we saw in the earlier figure reducing dimensions makes it easier to visualize data and reveal hidden patterns.
- **Prevent Overfitting**: With few features models are less likely to memorize the training data and overfit. This helps the model generalize better to new, unseen data improve its ability to make accurate predictions.

Disadvantages of Dimensionality Reduction

- **Data Loss & Reduced Accuracy:** Some important information may be lost during dimensionality reduction and affect model performance.
- **Choosing the Right Components:** Deciding how many dimensions to keep is difficult as keeping too few may lose valuable information while keeping too many can led to overfitting.

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Clustering methods

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Clustering

Clustering is an unsupervised learning task that aims to partition data into groups such that points within the same group are more similar to each other than to points in different groups, according to a chosen similarity or distance measure.

Jain, A. K., Murty, M. N., & Flynn, P. J. (1999). Hastie, T., Tibshirani, R., & Friedman, J. (2009).

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Main clustering models

Hierarchical

- Agglomerative (single, complete, average, Ward)
- Divisive

Builds a hierarchy of clusters (dendrogram).

Partition-based

- k-means
- k-medoids
- Mini-batch k-means

Fast and simple; assumes compact, spherical clusters.

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Main clustering models

Density-based

- DBSCAN
- HDBSCAN
- OPTICS

Clusters = dense regions; robust to noise.

Model-based (Probabilistic)

- Gaussian Mixture Models (GMMs)
- Bayesian mixture models

Assumes data generated from probability mixtures.

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Main clustering models

Spectral / Graph-based

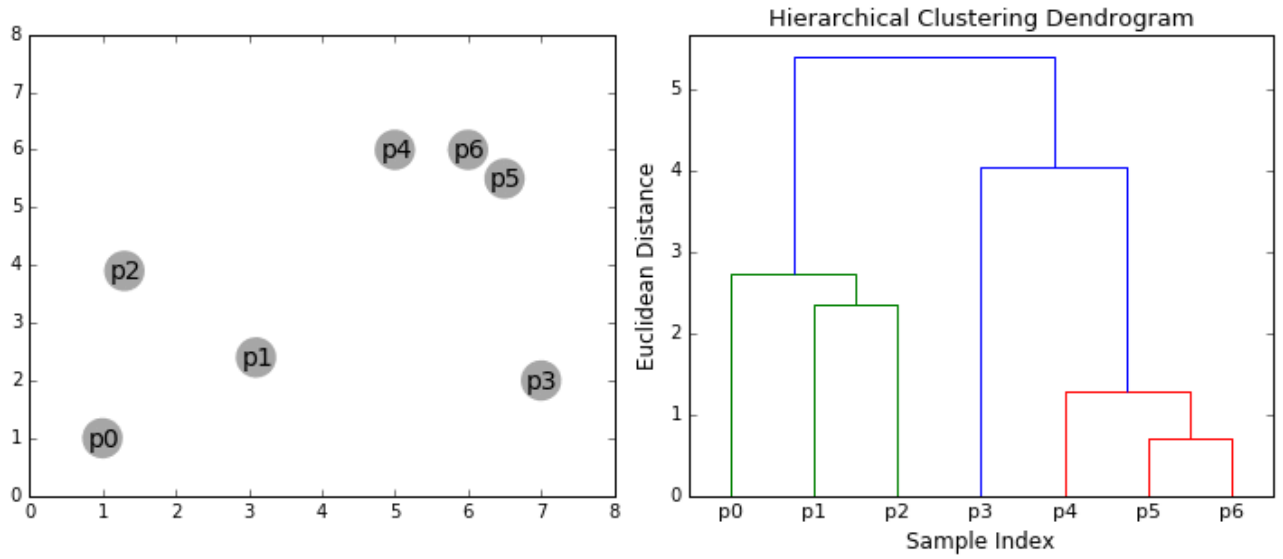
- Spectral clustering
- Normalized cuts

Uses eigenvectors of similarity graphs.

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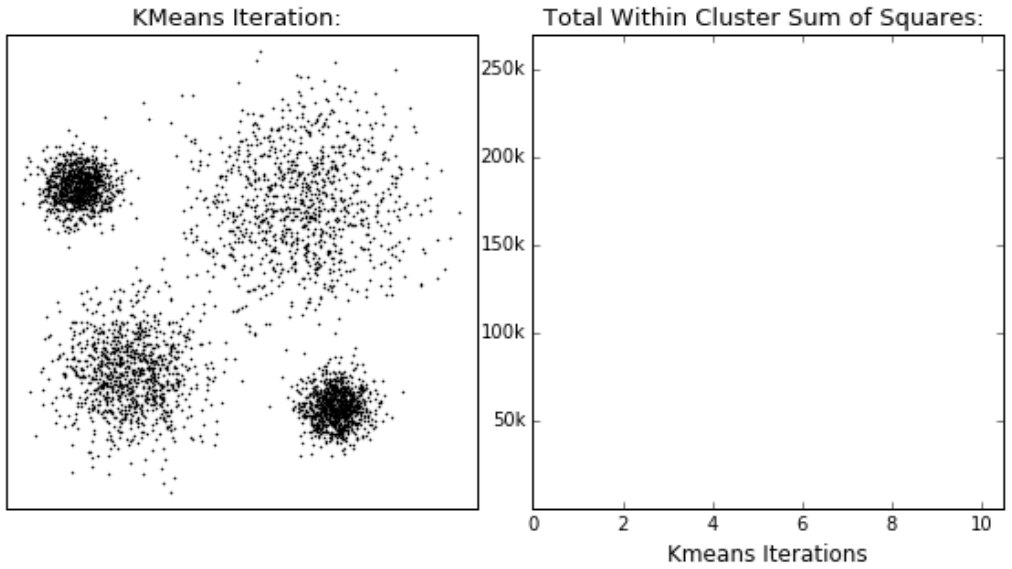
Hierarchical clustering

Hierarchical clustering is an unsupervised learning approach that builds a hierarchy of nested clusters by successively merging or splitting data points based on their pairwise similarities, producing a tree structure (dendrogram) that represents cluster relationships at multiple levels of granularity.



K-means and related methods

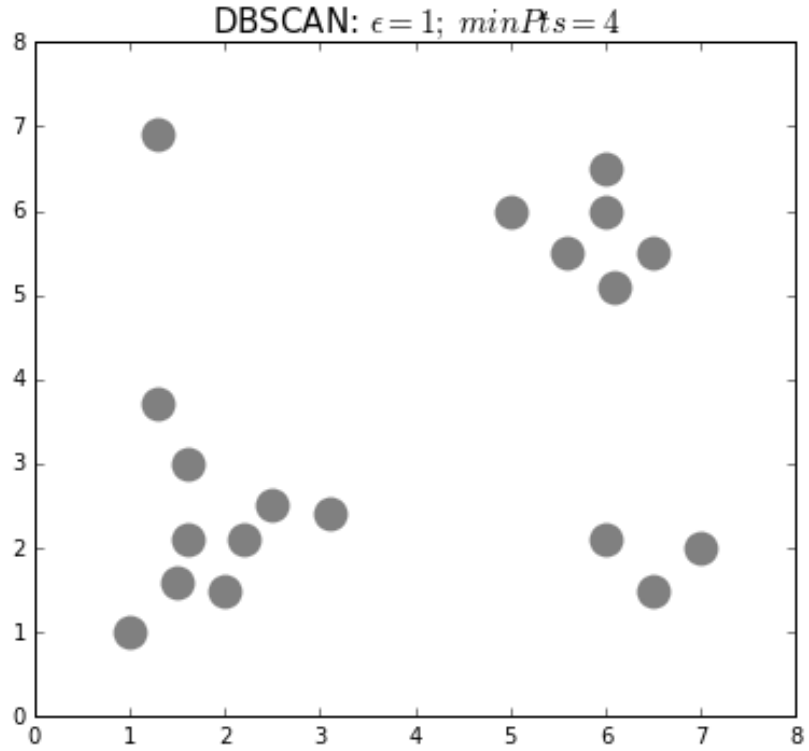
K-means and related centroid-based clustering methods aim to partition data into a fixed number of clusters by iteratively assigning each data point to the nearest cluster representative (centroid or medoid) and updating those representatives so as to minimize within-cluster dispersion, typically measured as the sum of squared distances to the cluster centers.



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DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups data points based on their proximity and density. The methodology involves randomly pick points which could start a cluster depending on the density of data around it, then, expands the cluster by adding all reachable points within some neighborhood ϵ . The steps are repeated for unvisited points until all points are processed.



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Clustering evaluation

Selecting the optimal number of clusters is a critical step in any unsupervised learning process and can also enhance supervised learning by revealing significant group structures within the data. In this section, we review two methods for determining the ideal number of clusters: the Elbow method and the Silhouette score.

Elbow

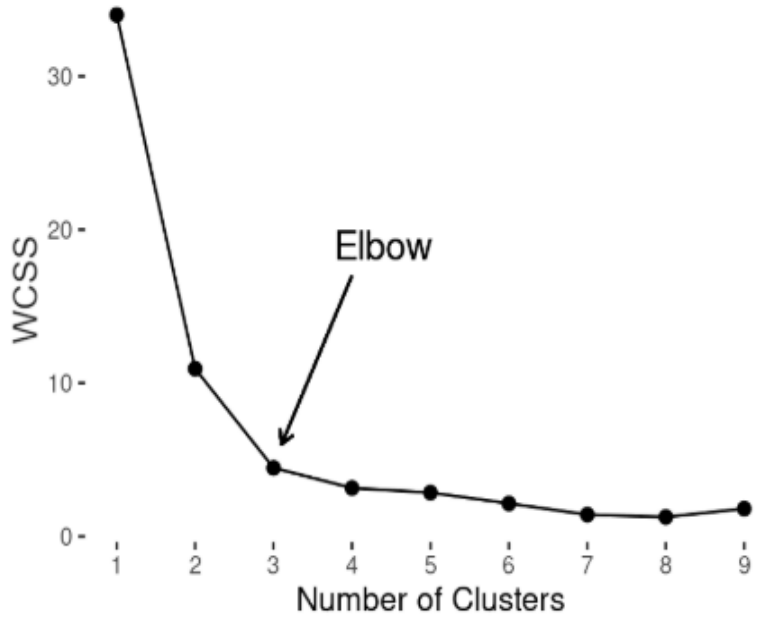
The Elbow Method is a popular technique used for this purpose in K-Means clustering and tipically consist in the following steps:

- For a specific value of k calculate Within-Cluster Sum of Squares (WCSS). This assess how spread out the data points are within each cluster.

For a given k, the WCSS is:

$$\text{WCSS} = \sum_{i=1}^{k} \sum_{y \in C_i} [y - \mu_i]^2$$

where C_i is the i-th cluster set ($i = 1, \ldots, k$), y is one of its observation ($y \in C_i$) and μ_i is it centroid.



- Plot a graph with k on the X-axis and WCSS(k) on the Y-axis.Based on the plot, we identify the Elbow Point.

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Silhoutte score

Silhouette score is a metric used to evaluate how good clustering results are in data clustering.
This score is calculated by measuring each data point's similarity to the cluster it belongs to and how different it is from other clusters (within and between distance).

- For each data point, the average distance (a_i) to other data points within the same cluster is calculated. This value represents **the similarity level of the data point to others in its cluster**.
- For each data point, the average distance (b_iC) to all other clusters it doesn't belong to is computed. This value indicates **how different the data point is from data points in other clusters**.
- From this, we compute b_i as the min b_iC. This value indicates **the cluster C as the "neighbor" of data point i, (or its second best option).**

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Silhoutte score

- The cluster Silhouette score is calculated using the formula

$$\mathrm{CSS}(i) = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1,1]$$

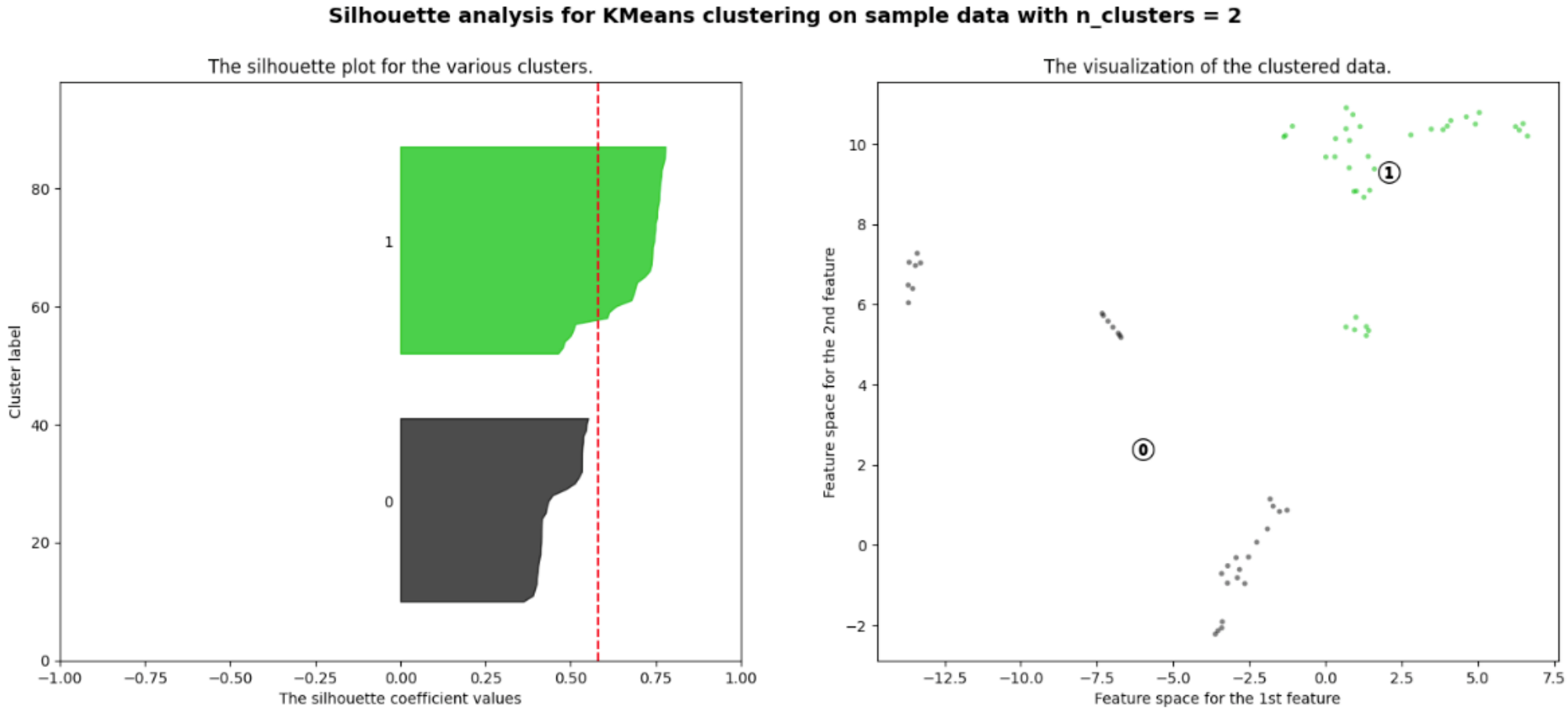
- By taking the average of the Silhouette scores calculated for each data point, an overall Silhouette score is obtained, which measures the success of clustering results.

$$\text{Silhouette Score (SS)} = \frac{\sum_{i=i}^{k} CSS(i)}{k}$$

SS	Interpretation
0.71 to 1	A strong structure has been found
0.51 to 0.7	A reasonable structure has been found
0.26 to 0.5	The structure is weak and could be artificial
≤ 0.25	No substantial structure has been found

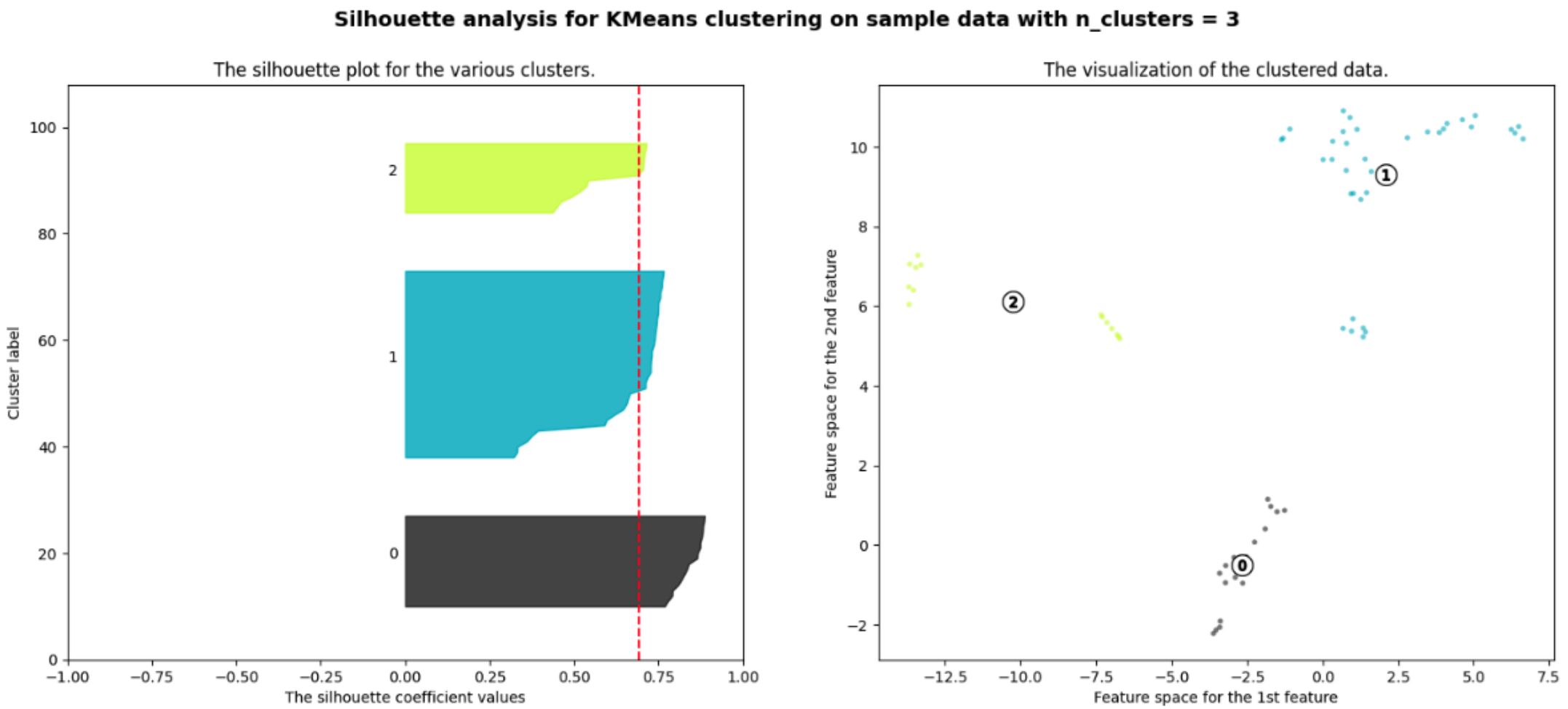
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Silhoutte score



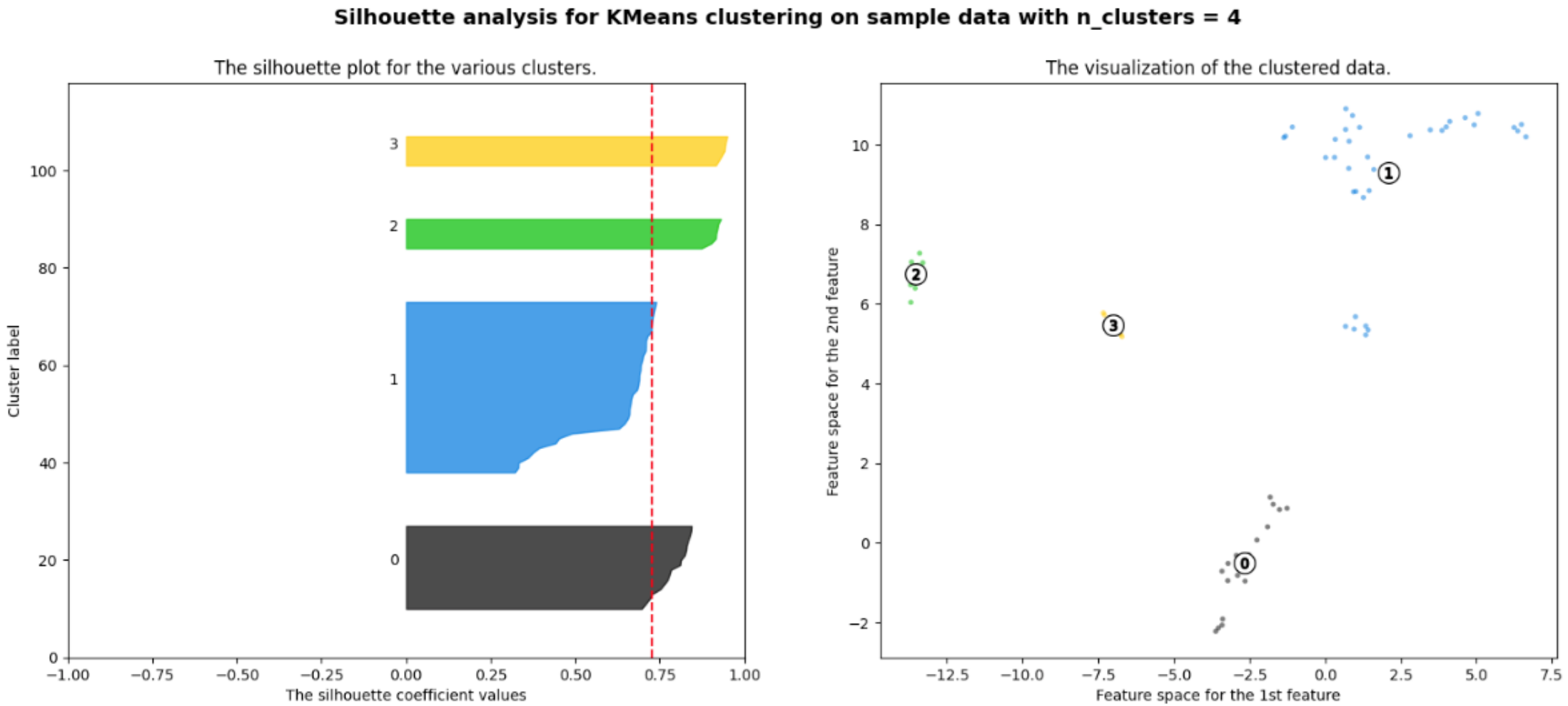
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Silhoutte score



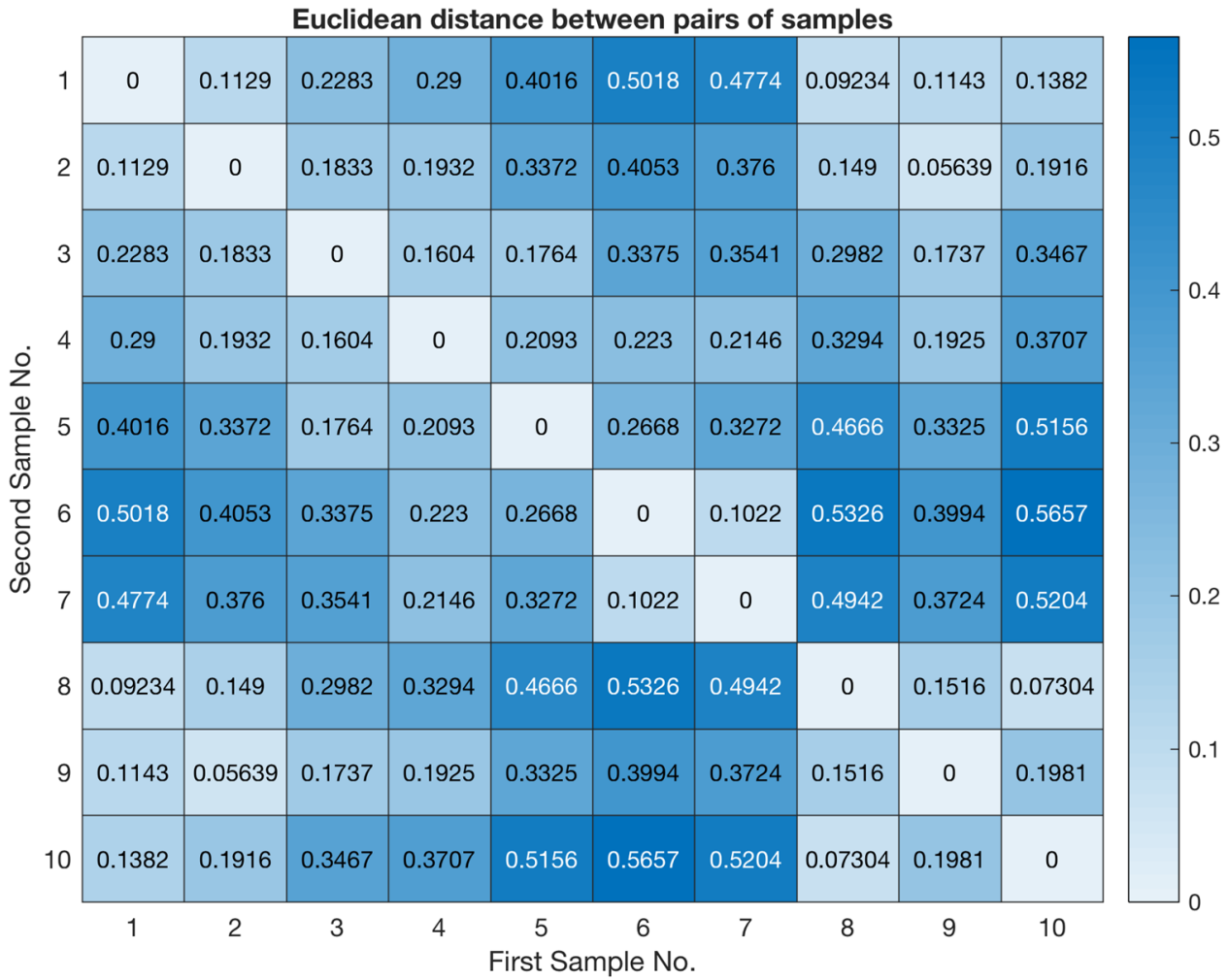
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Silhoutte score



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Clustering based on distance matrices



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Clustering based on distance matrices

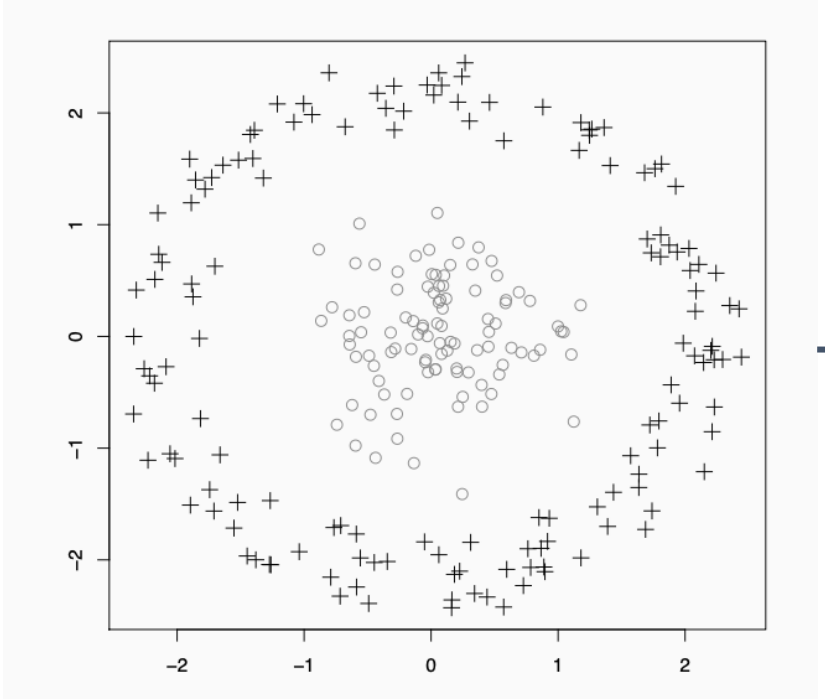
- Cons: lost information and maybe, interpretation about the original variables
- Pros: you can use any distance metric (even rank-based) as long as it meets those properties:

  - **Non-negativity:** $d(x, y) \geq 0$
  - **Identity:** $d(x, y) = 0 \iff x = y$
  - **Symmetry:** $d(x, y) = d(y, x)$
  - **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$

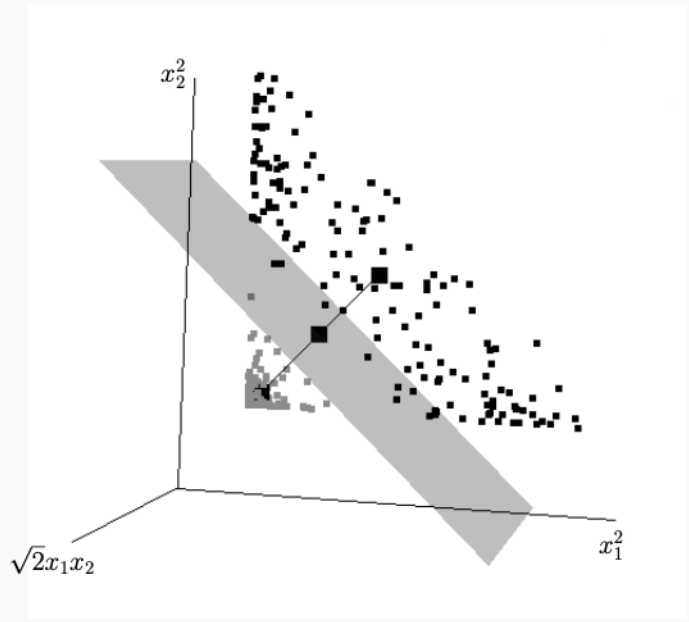
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Clustering based on distance matrices

Example: Kernel methods



$$\phi : \mathbf{x} = (x_1, x_2) \mapsto \phi(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1x_2);$$



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Example: Kernel methods

A **kernel function** is a function that measures the similarity between two observations and can be interpreted as an inner product in a (possibly high-dimensional) feature space:

Kernel trick

$$K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$$

where:

- $\mathbf{x}$ and $\mathbf{z}$ are input samples,
- $\phi(\cdot)$ is a feature mapping,
- $\langle \cdot, \cdot \rangle$ denotes the inner product.

kernel matrix (**Gram**) is n x n matrix containing all pairwise kernel evaluations

$$\mathbf{K} = \begin{bmatrix} K(\mathbf{x}_1, \mathbf{x}_1) & K(\mathbf{x}_1, \mathbf{x}_2) & \cdots & K(\mathbf{x}_1, \mathbf{x}_n) \\ K(\mathbf{x}_2, \mathbf{x}_1) & K(\mathbf{x}_2, \mathbf{x}_2) & \cdots & K(\mathbf{x}_2, \mathbf{x}_n) \\ \vdots & \vdots & \ddots & \vdots \\ K(\mathbf{x}_n, \mathbf{x}_1) & K(\mathbf{x}_n, \mathbf{x}_2) & \cdots & K(\mathbf{x}_n, \mathbf{x}_n) \end{bmatrix}$$

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Classification models

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- High dimensional data (but, very high!)
- Small *n*, big *d*
- Unbalanced categories
- Overfitting

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Given a set of training **labeled** data

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}; \quad \mathbf{x} \in \mathbb{R}^d,$$

The objective is to learn a function for prediction:

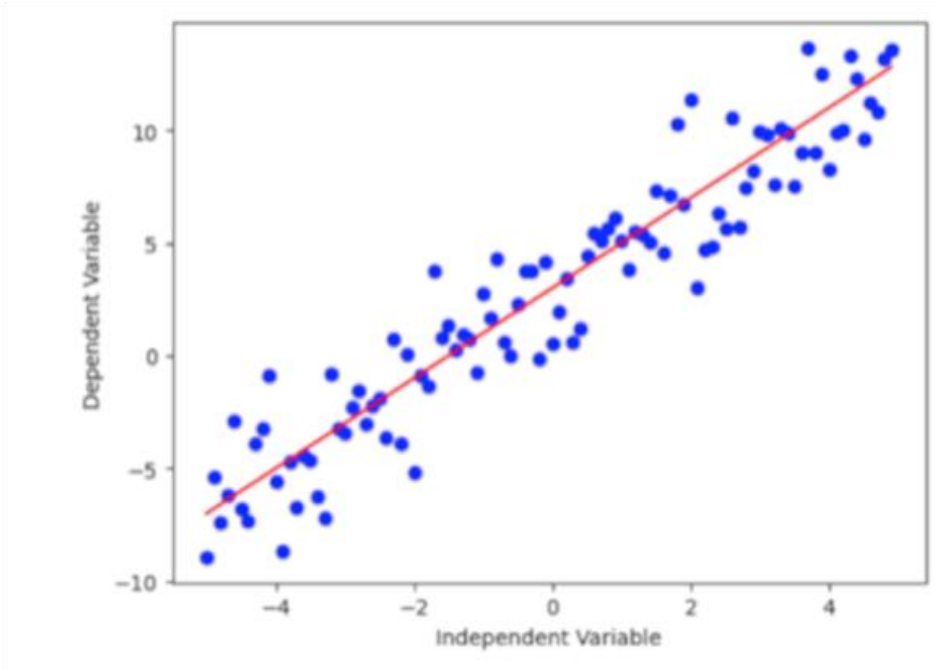
$$y = f(\mathbf{x}; \boldsymbol{\theta}).$$

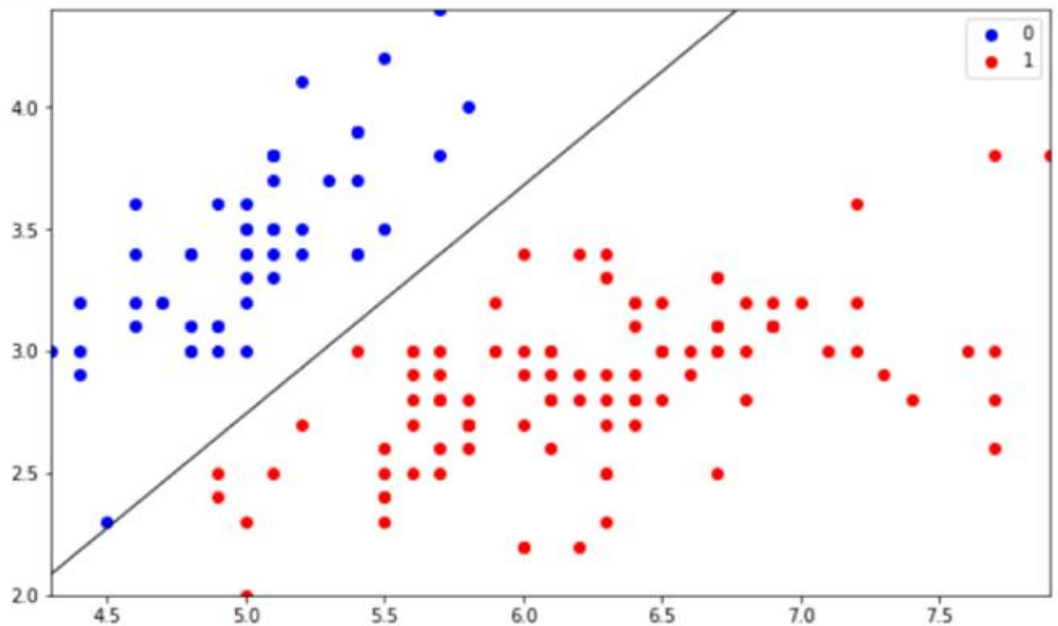
Regression: $f : \mathbb{R}^d \mapsto \mathbb{R}$

Classification: $f : \mathbb{R}^d \mapsto \mathbb{Z}$

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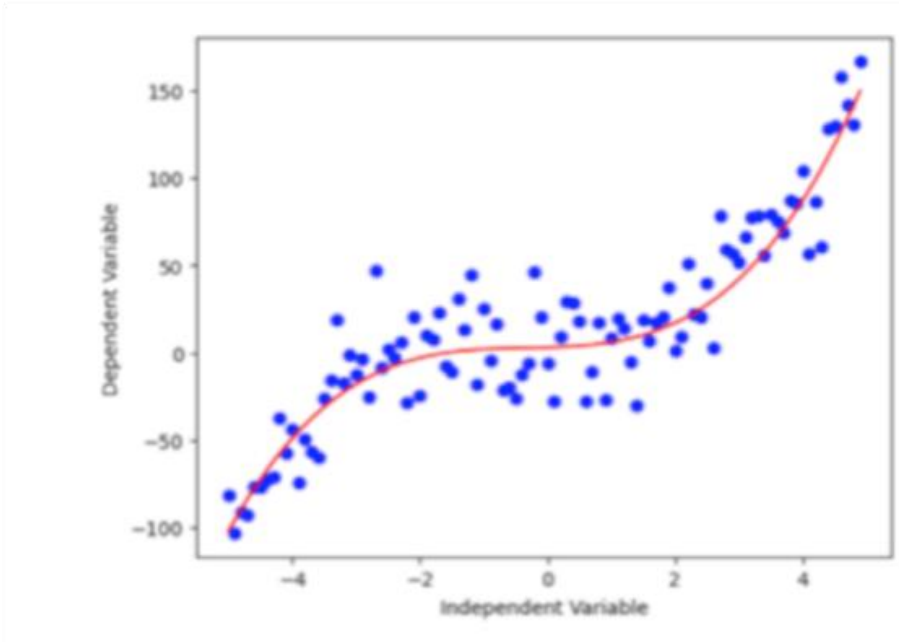
f can be simple:

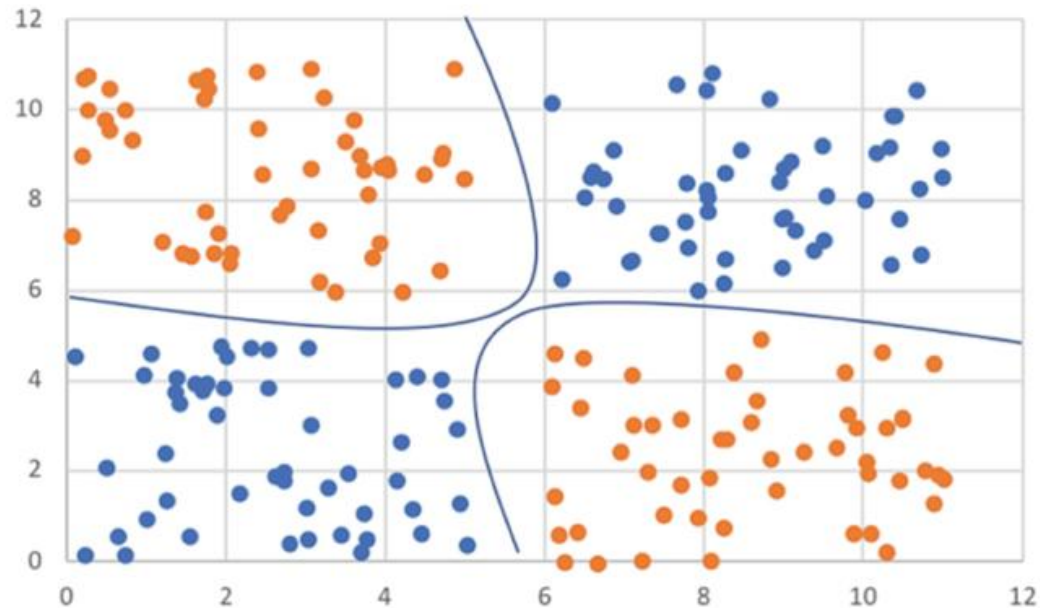




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Or more complex:





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We will focus on classification problems, where $y \in \{1,\ldots,K\}$ categories, and the objective is to find a learner

$$f : \mathbb{R}^d \mapsto \{1,\ldots,K\}$$

If, for each $y_k \in \{1,\ldots,K\}$, we can obtain $P(y = y_k \,|\, \mathbf{x})$ and $\sum_{k=1}^{K} P(y = y_k \,|\, \mathbf{x}) = 1$,

then, our best estimate of *y* is

$$\hat{y} = \arg\max_{y_k} P(y = y_k \,|\, \mathbf{x})$$

Which is called the Bayes optimal classifier.

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In supervised ML, f is obtained by **minimizing a loss function**.

- For regression, we can use MSE: $\frac{1}{n}\sum_{i=1}^{n}(y_i - f(\mathbf{x}_i))^2$ or some related loss functions (MAE, RMSE).
- For classification, we have several options depending on the model:

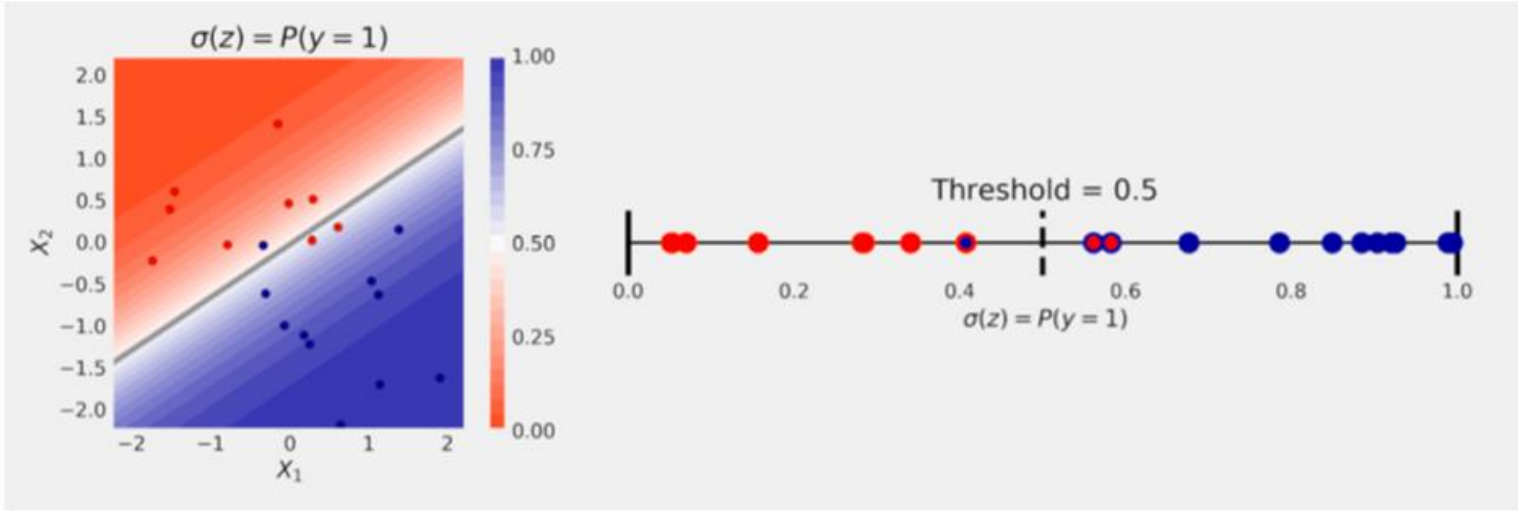
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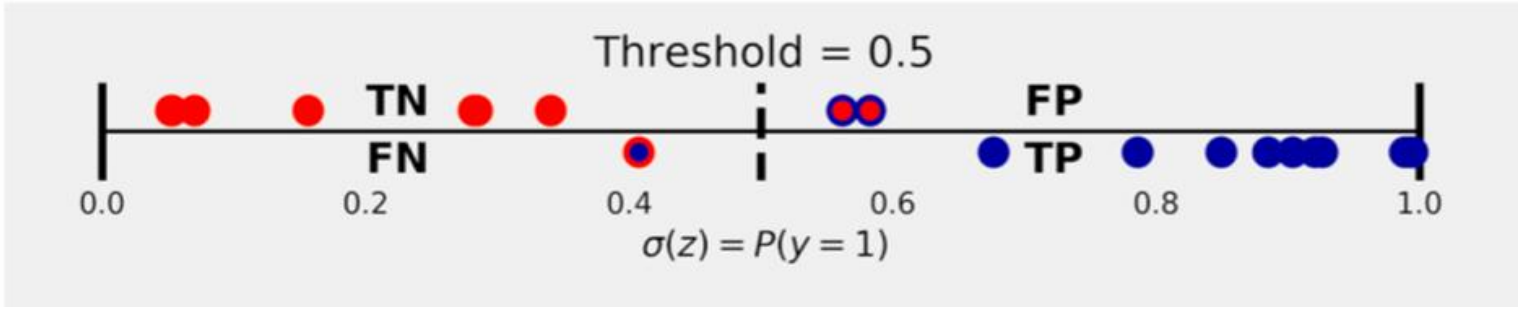
Loss Function	Typical Use	Formula	Key Idea
Binary Cross-Entropy (Log Loss)	Binary classification	$-[y \log(p) + (1-y) \log(1-p)]$	Measures the discrepancy between predicted and true class probabilities.
Categorical Cross-Entropy	Multiclass classification	$-\sum_{i=1}^{K} y_i \log(p_i)$	Generalizes cross-entropy to multiple classes.
Hinge Loss	Support Vector Machines (SVMs)	$\max(0, 1 - yf(x))$	Encourages a large margin between classes while penalizing margin violations.
Exponential Loss	AdaBoost and boosting methods	$e^{-yf(x)}$	Assigns exponentially larger penalties to misclassified examples.

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Performance metrics:

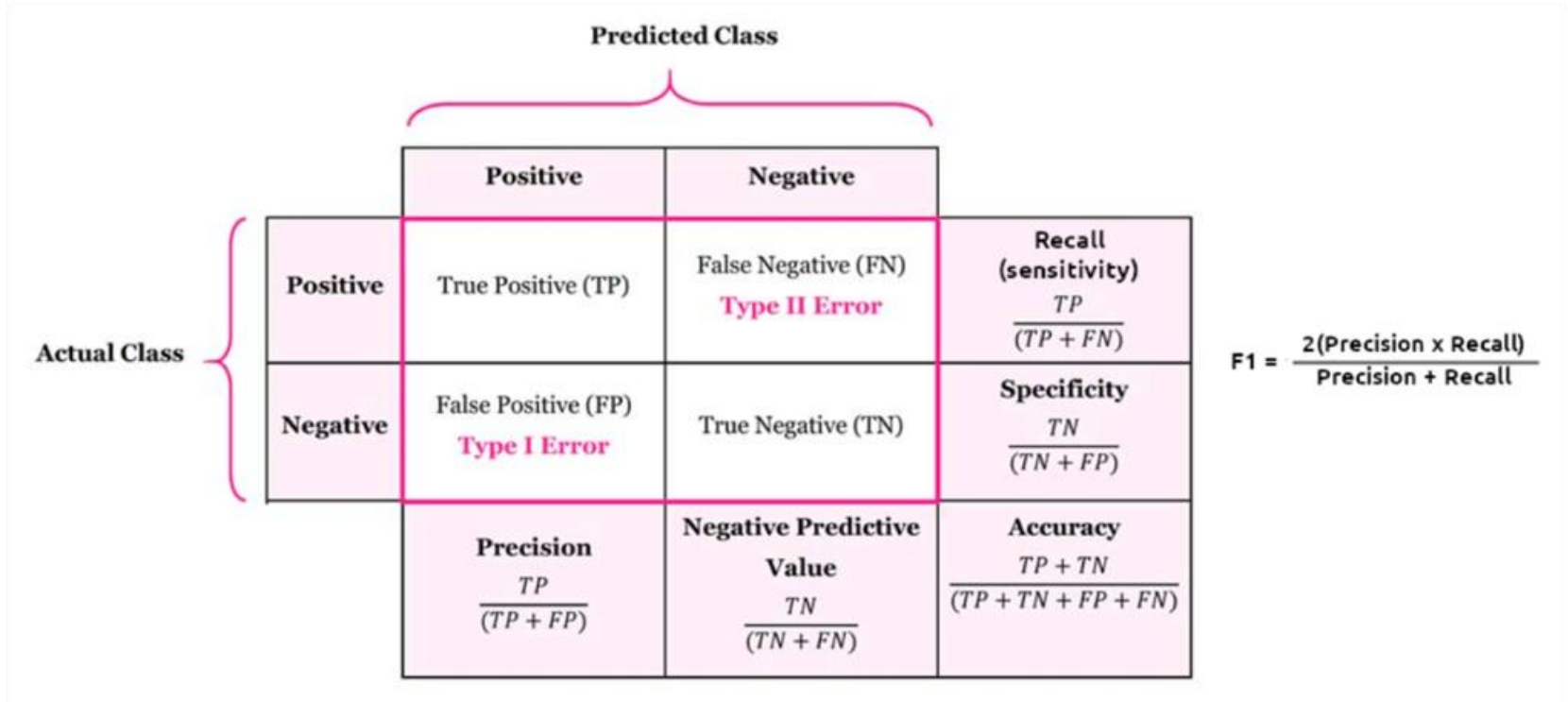
For binary classification





BiotrAIn

For binary classification

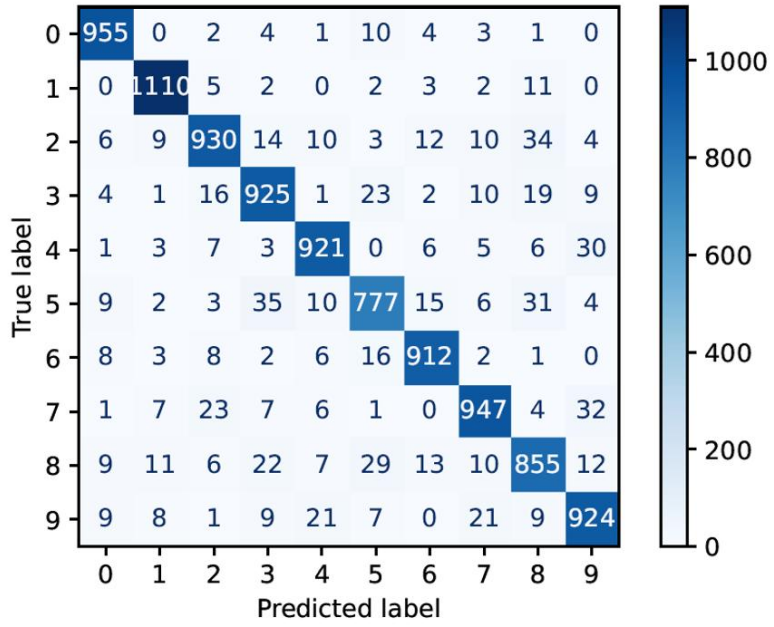


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Multiclass classification metrics

```
print(metrics.classification_report(y_test, y_pred2))
```

	precision	recall	f1-score	support
0	0.80	0.75	0.77	16
1	0.45	0.49	0.47	43
2	0.67	0.14	0.24	14
3	0.79	0.79	0.79	14
4	0.56	0.56	0.56	16
5	0.00	0.00	0.00	22
6	0.70	0.76	0.73	101
7	0.85	0.94	0.89	228
8	0.68	0.97	0.80	70
9	0.90	0.83	0.86	86
10	0.98	1.00	0.99	87
11	0.62	0.29	0.40	34
12	0.73	0.76	0.74	21
13	0.45	0.25	0.32	20
14	0.25	0.20	0.22	20
accuracy			0.77	792
macro avg	0.63	0.58	0.59	792
weighted avg	0.74	0.77	0.74	792



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ROC Curve (Receiver Operating Characteristic)

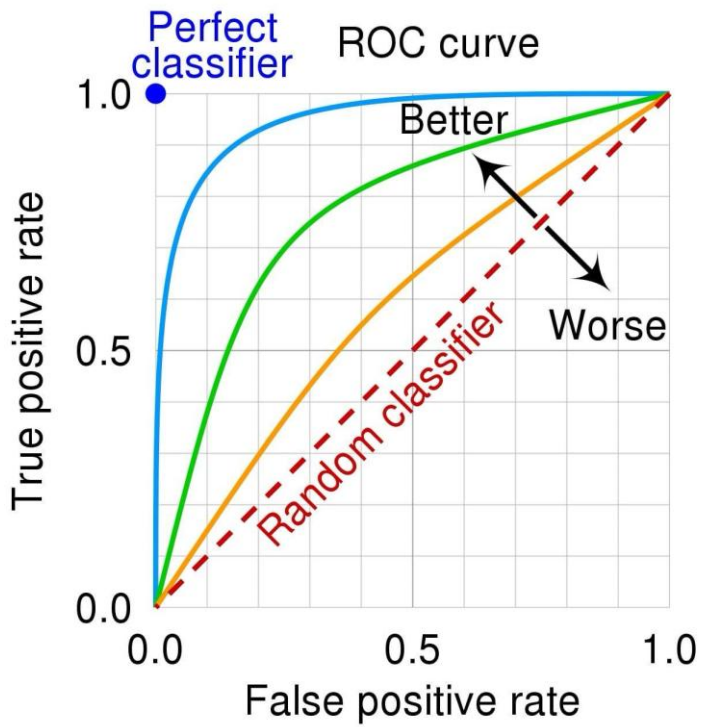
The **ROC curve** is a graphical tool used to evaluate the performance of a binary classifier across **all possible decision thresholds**.

- **X-axis:** False Positive Rate (FPR)

$$\text{FPR} = \frac{FP}{FP + TN}$$

- **Y-axis:** True Positive Rate (TPR), also known as **Recall** or **Sensitivity**

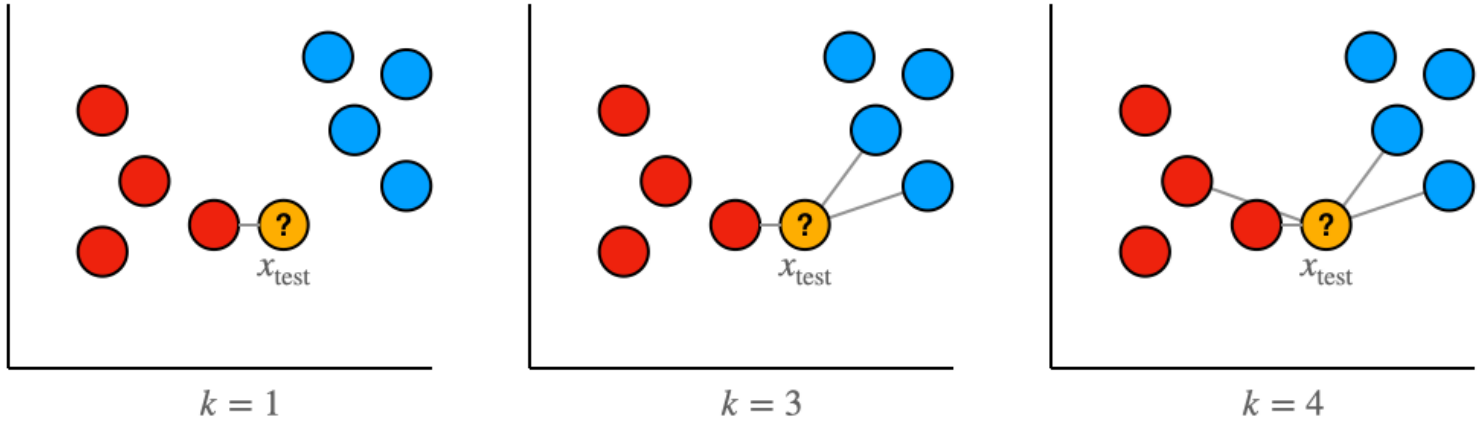
$$\text{TPR} = \frac{TP}{TP + FN}$$



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Classification models

k-Nearest Neighbors (KNN) is a simple, non-parametric classification algorithm that assigns a class to a new sample based on the majority class among its *k* closest training examples in the feature space. It is intuitive, easy to implement, and effective for many pattern recognition tasks.

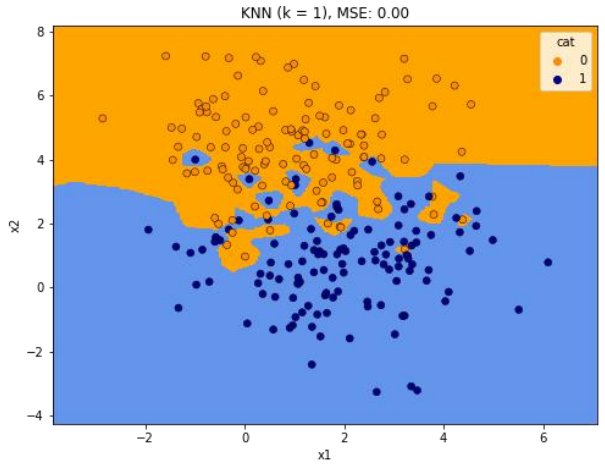


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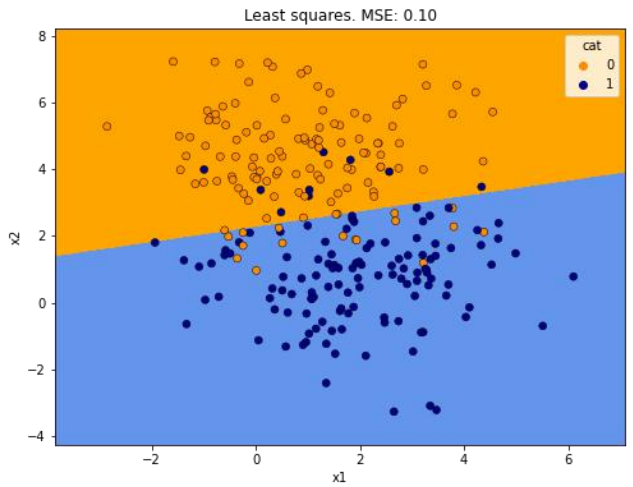
Classification models

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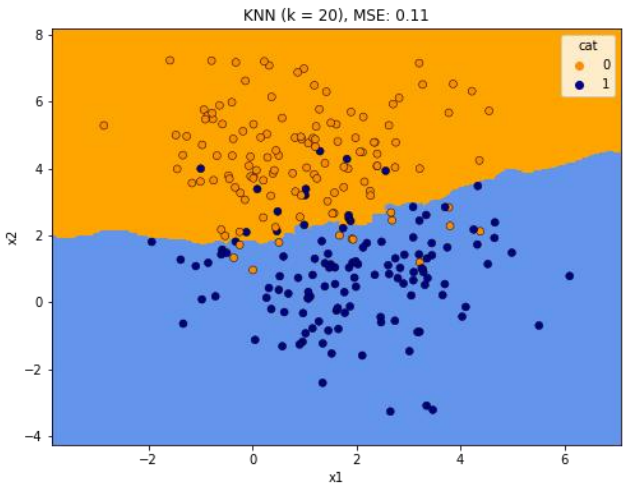
K=1



K=100



K=20

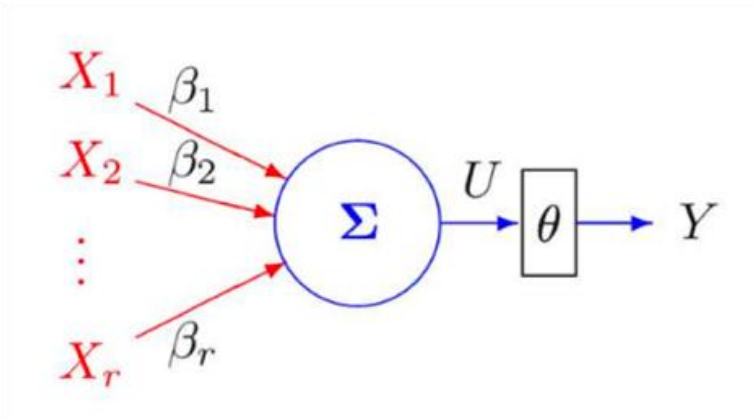


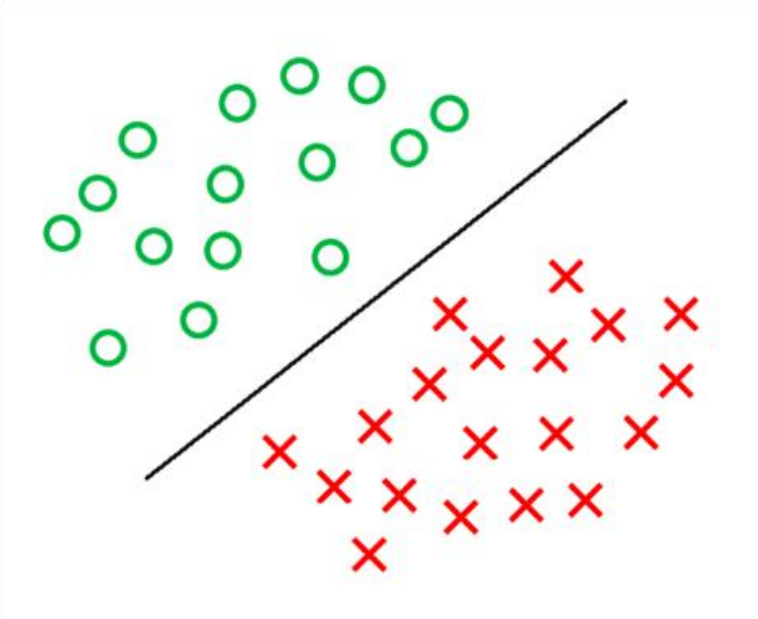
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Classification models

We start with a simple linear model for classification

$$f(\mathbf{x}) = \mathbf{x}'\boldsymbol{\beta}$$



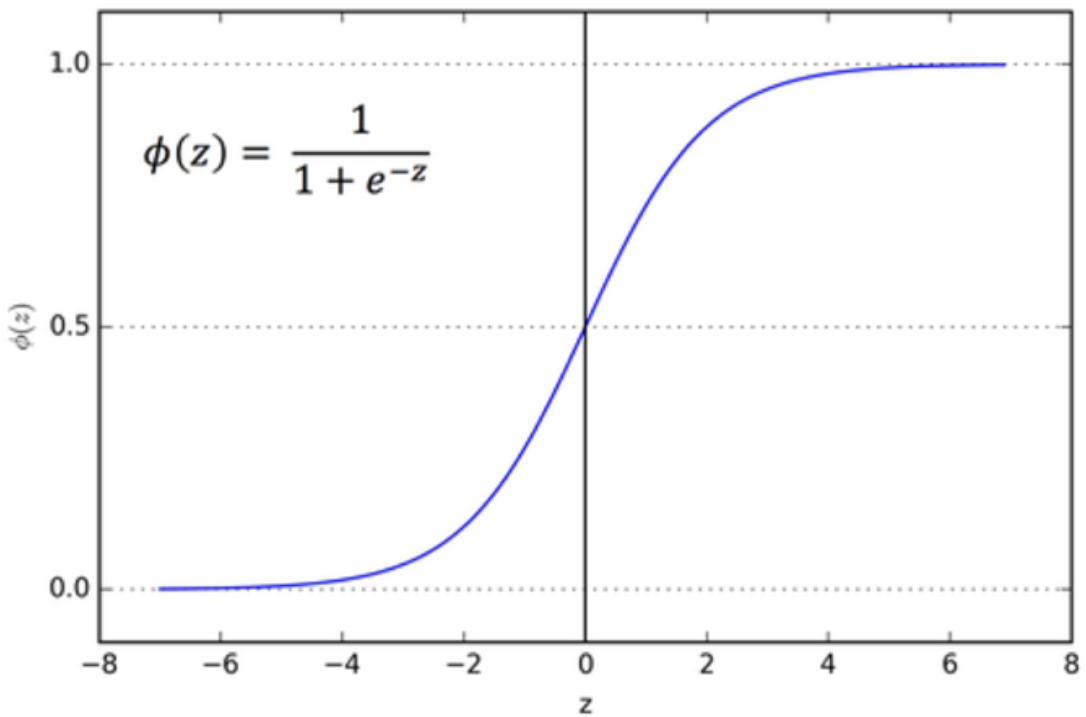


BiotrAIn

Classification models

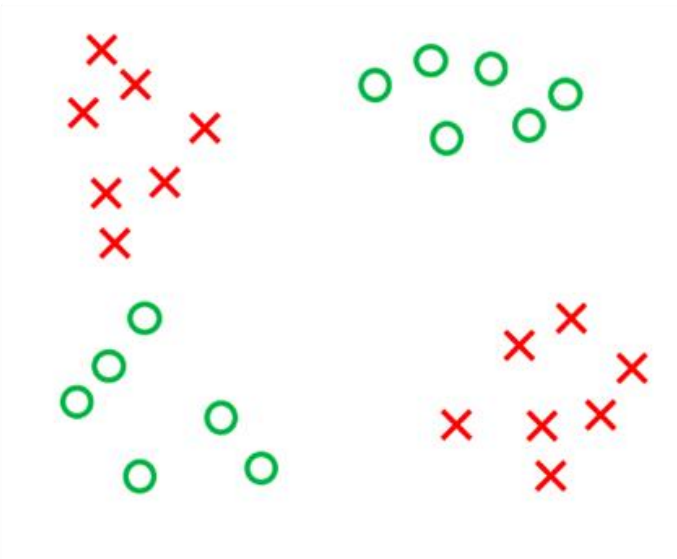
The perceptron model is a specific case of **logistic regression** when we use the so called sigmoid function to obtain posterior probabilities:

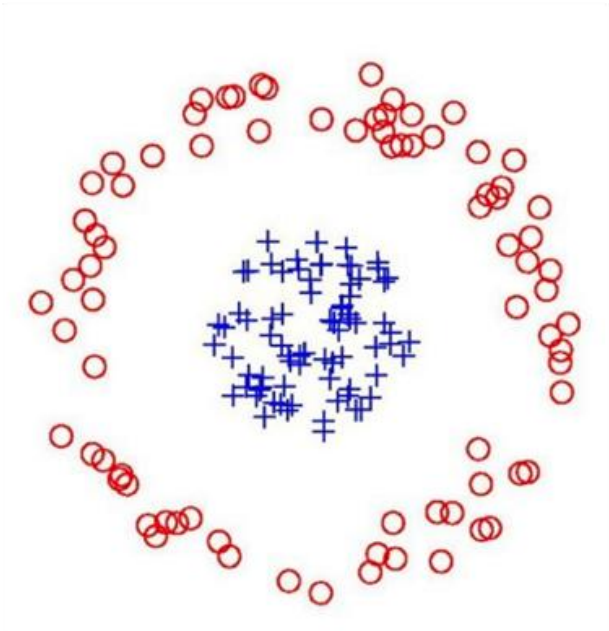
$$P(y = 1 \,|\, \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{x}'\beta}}$$



BiotrAIn

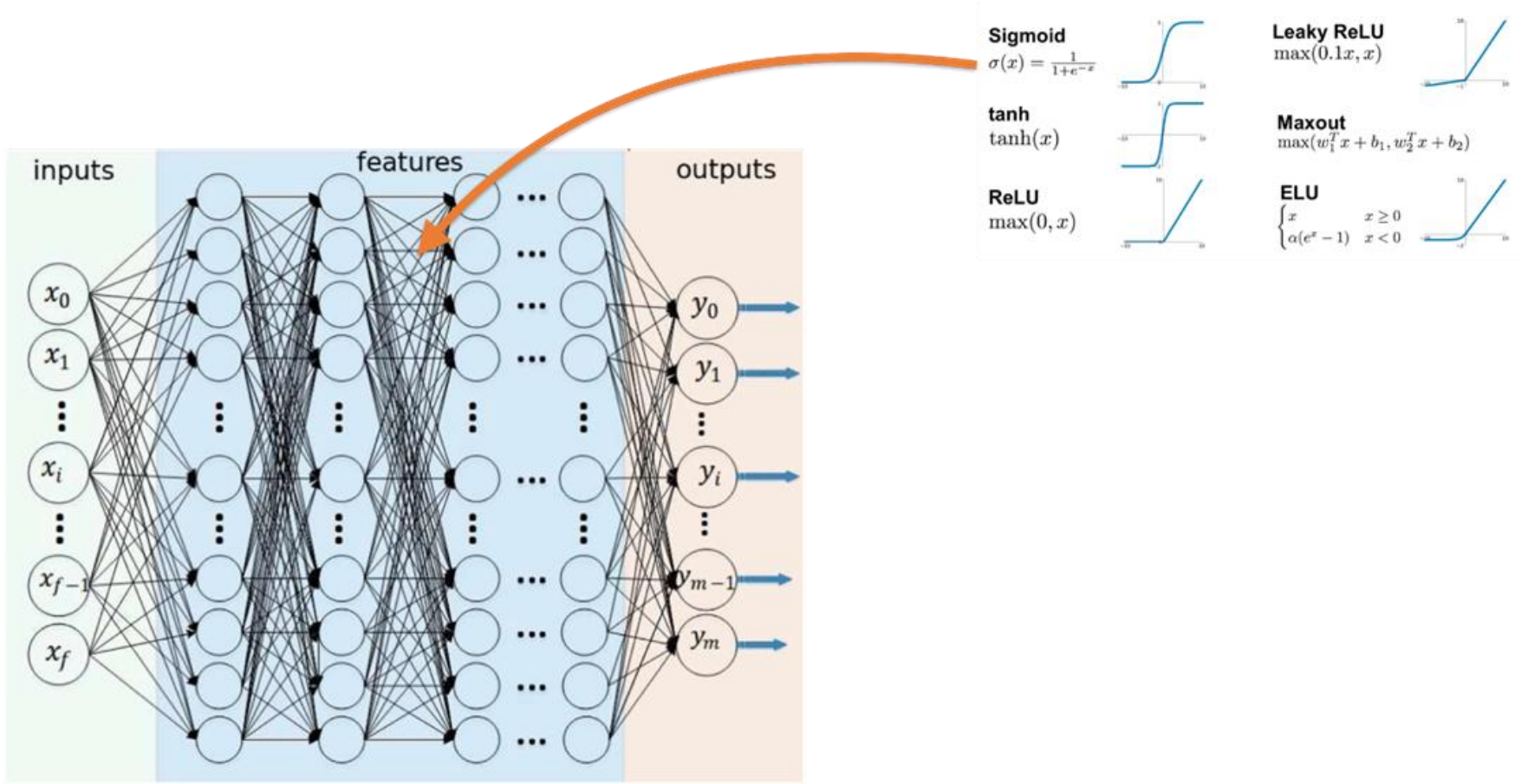
However, these models fail when our data are not linearly separable:





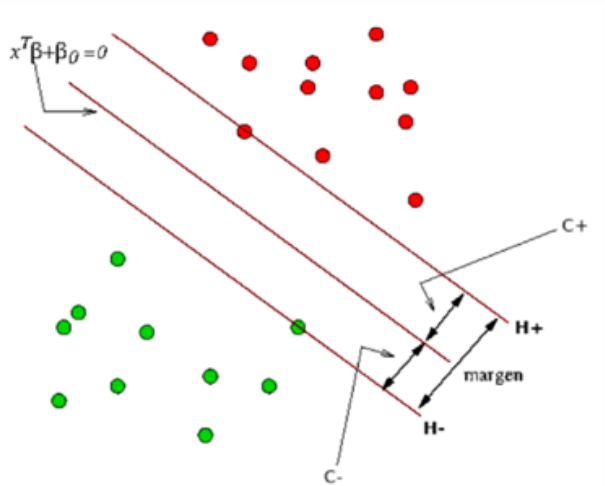
BiotrAIn

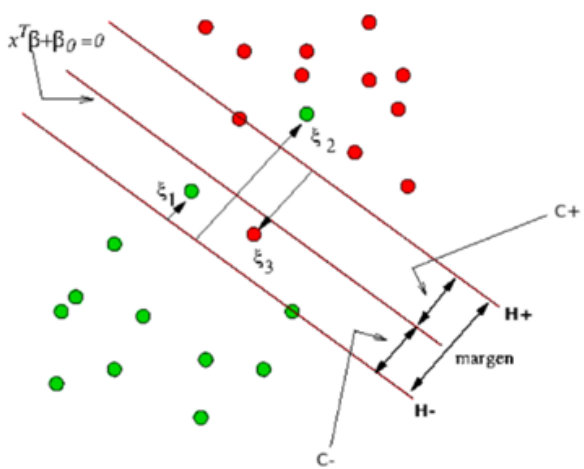
One solution for this problem, is to add layers of linear functions with non-linear transformations (**multi-layer perceptron or neural networks**):

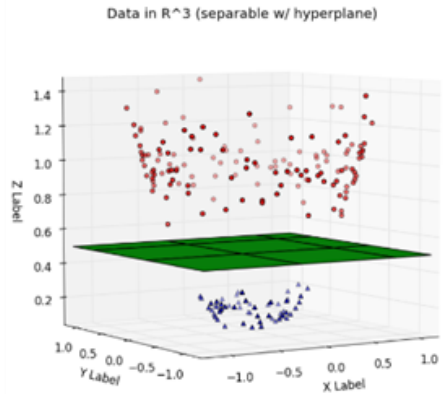


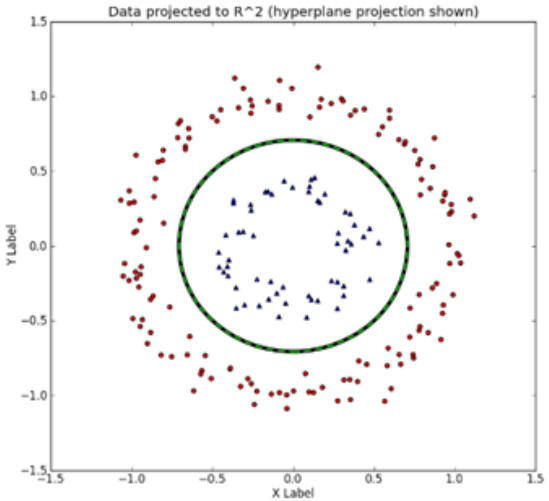
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Other idea is to find the optimal separating hyperplane and implicit variable transformations, such as **Support Vector Machines (SVM)**









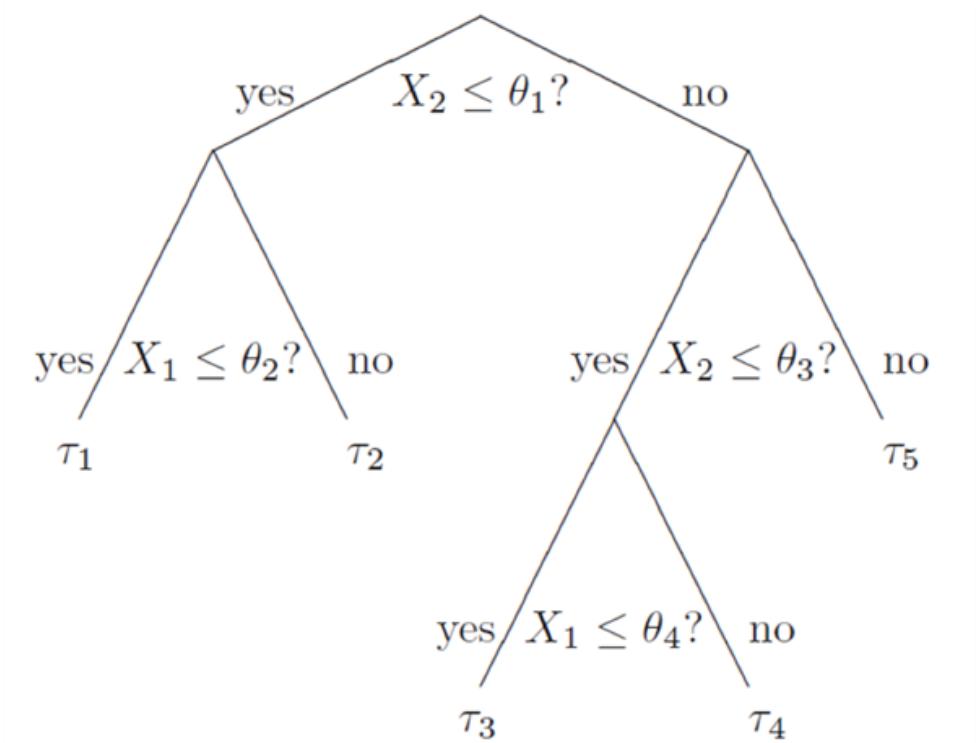
Separable case

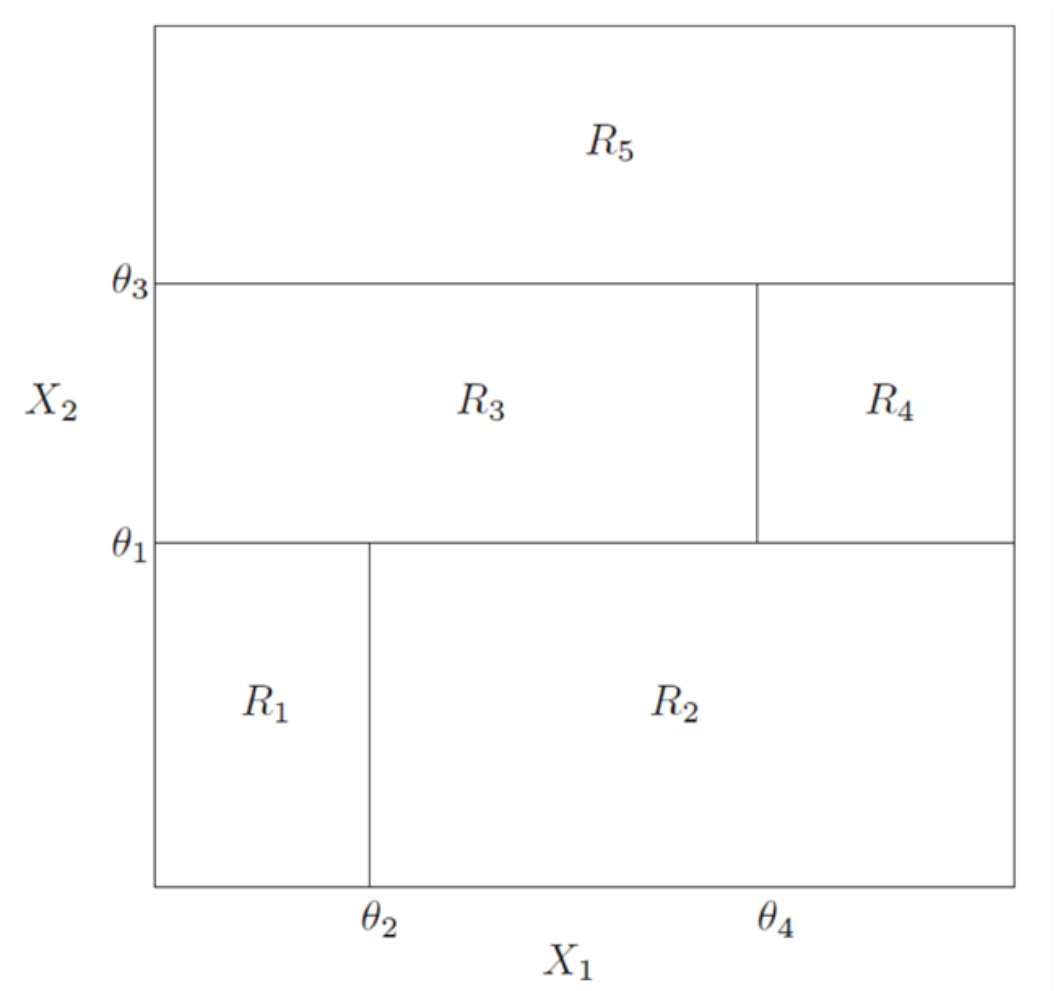
Non-separable case

Non-linear case (kernel trick)

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Classification and regression trees are another family of models which partition the input space





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A classification tree is a **decision tree** which **recursively splits the dataset** based on feature values to create a tree-like structure, where:

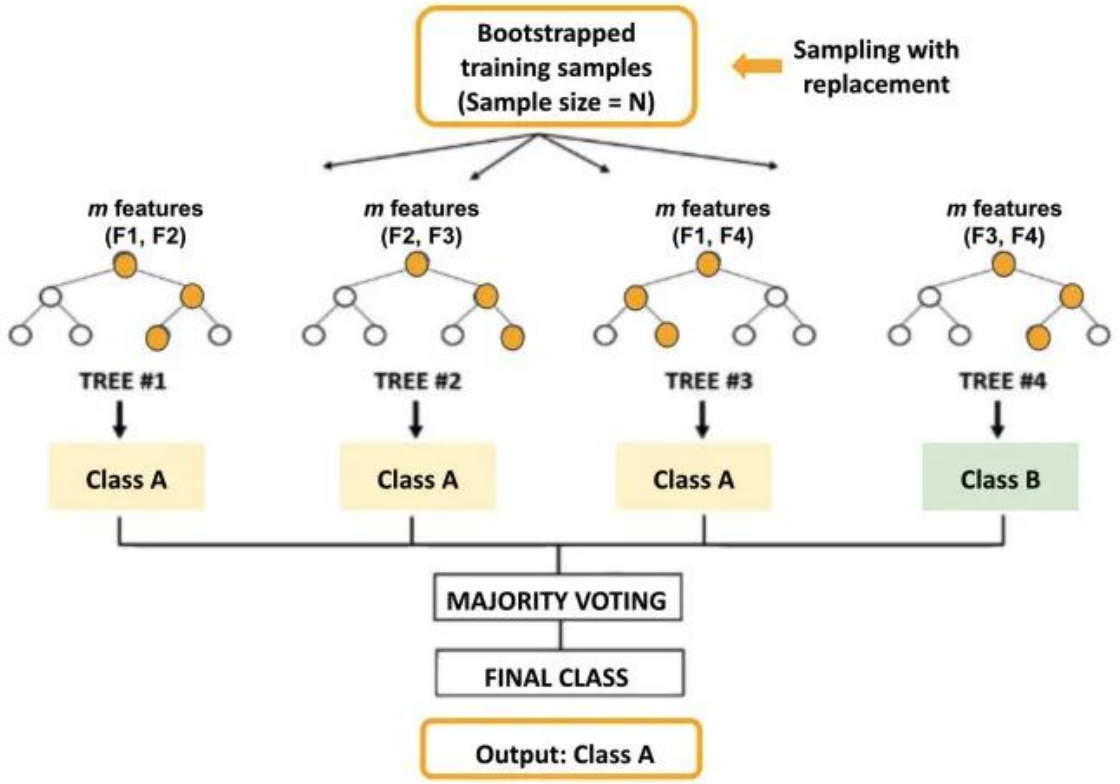
- Internal nodes represent decision rules based on feature values.
- Branches represent possible outcomes of a decision.
- Leaf nodes represent class labels (final predictions)

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Finally, ensemble models, such as **Boosting**, **Bagging** and **Random Forests**, are techniques that combine multiple individual models to improve overall performance, accuracy, and robustness. By leveraging the strengths of multiple models, ensemble methods help reduce overfitting and increase predictive power.

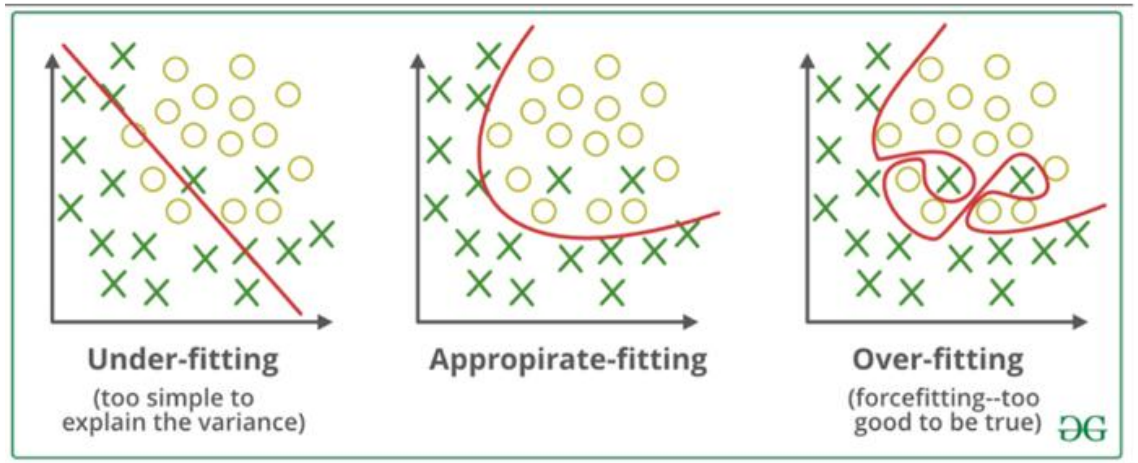
Training Data
(Sample size, N= 6, No. of features, F = 4)

F1	F2	F3	F4	Y
2.1	0	400	-9	A
3.0	1	890	-42	B
2.2	1	929	0	B
4.0	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A

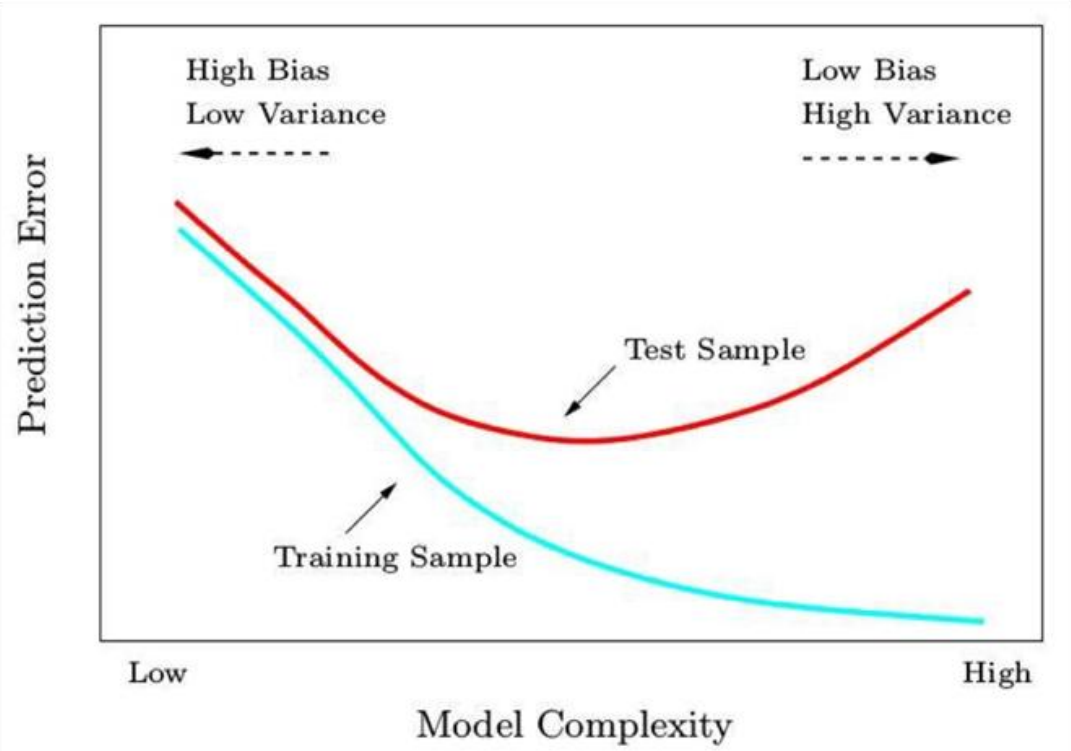


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An important aspect to consider when we choose a classification model, is the **overfitting**, which can be understood with the so-called **bias-variance tradeoff**:



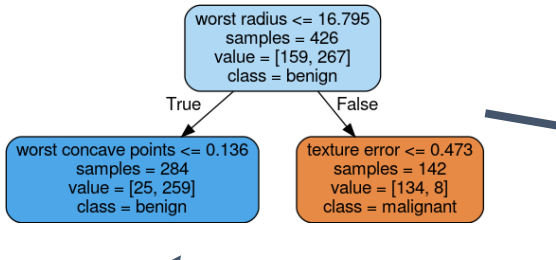
Source: https://medium.com/@pingsubhak/

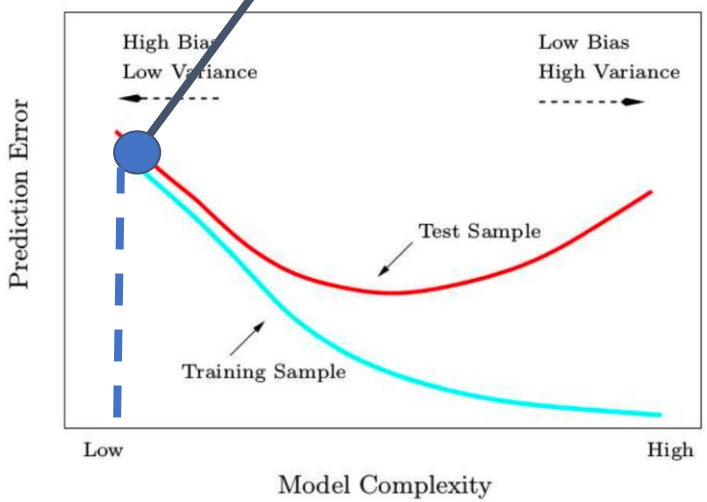


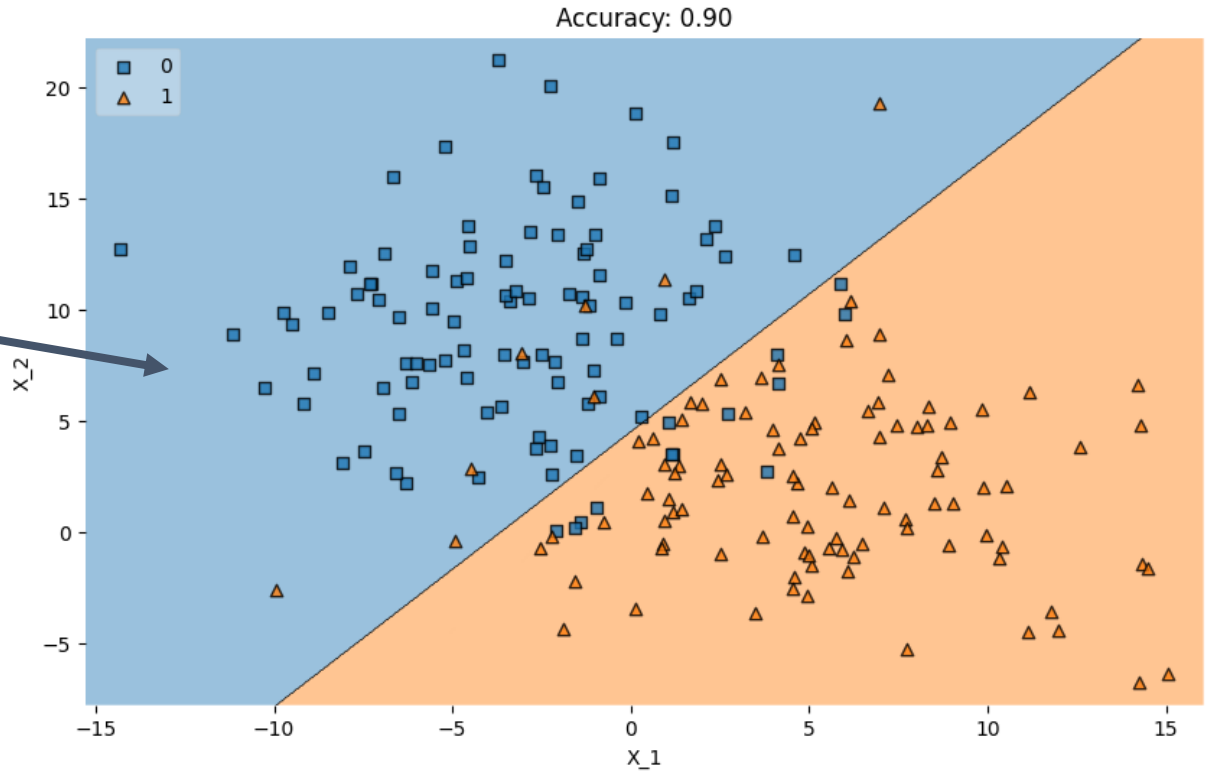
BiotrAIn

Bias-variance tradeoff

A simple model



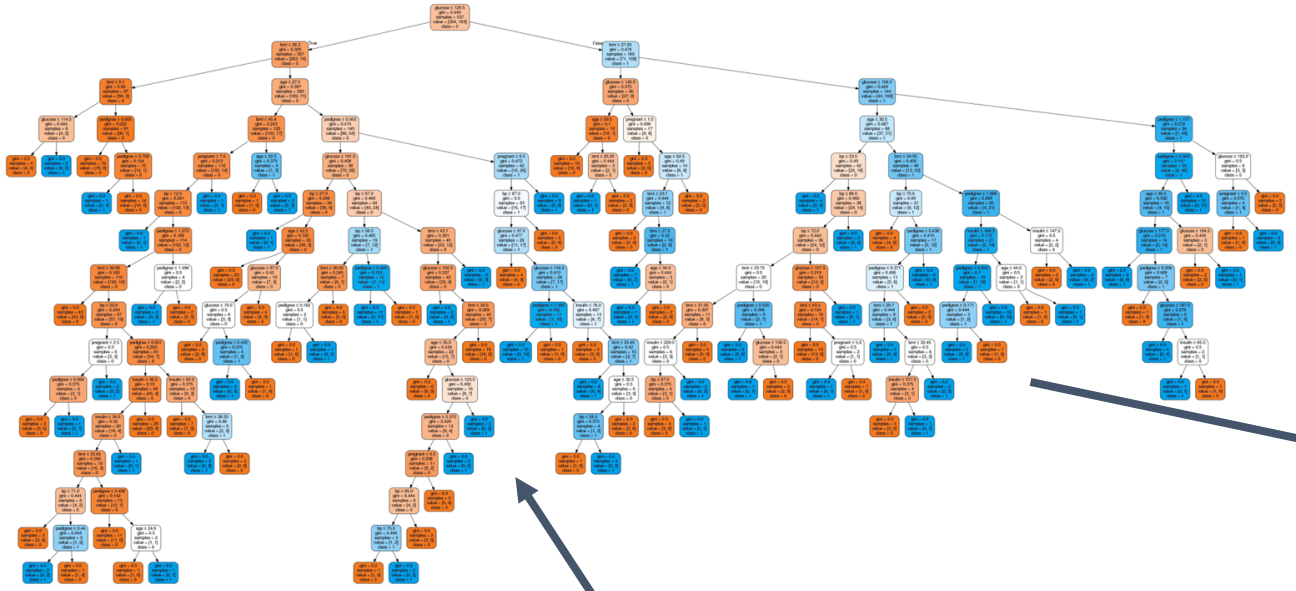


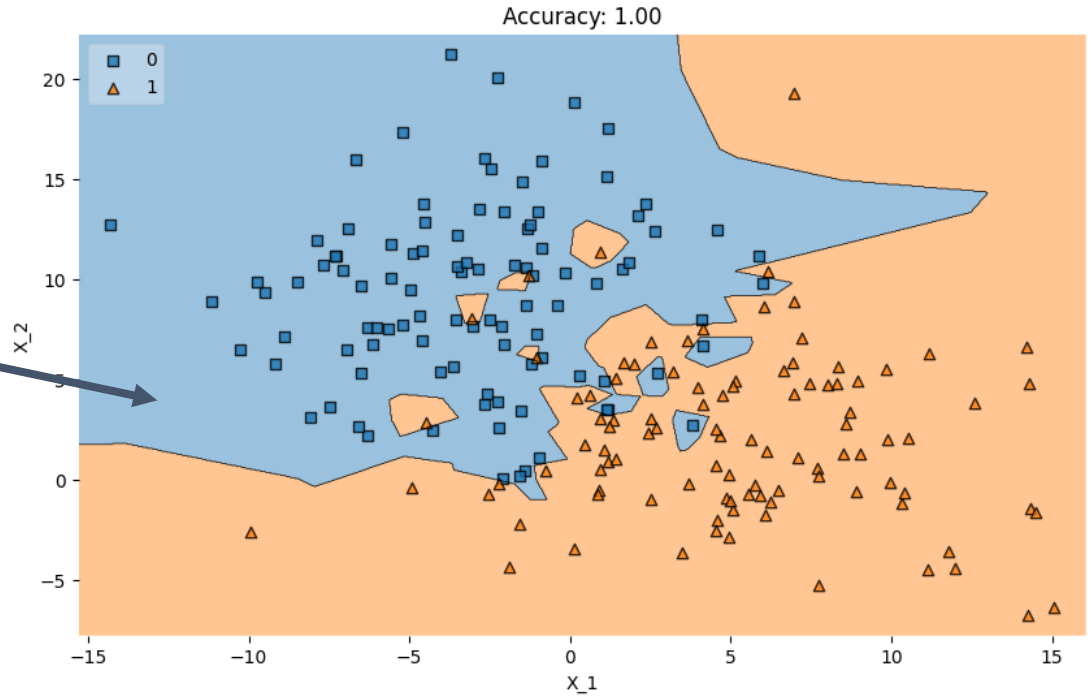


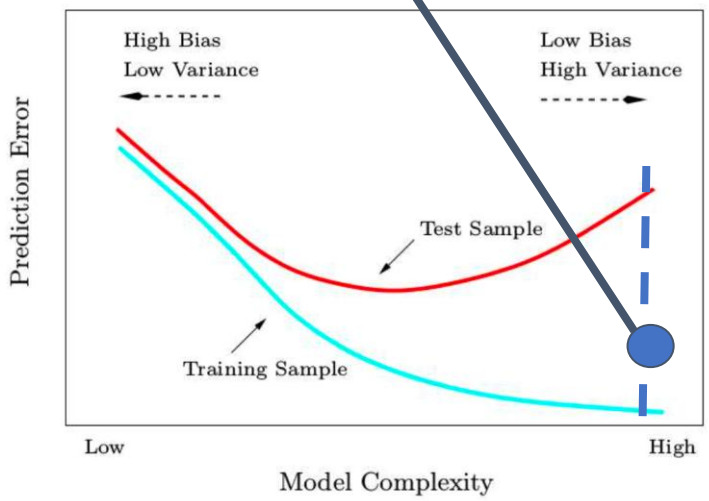
BiotrAIn

Bias-variance tradeoff

A complex model



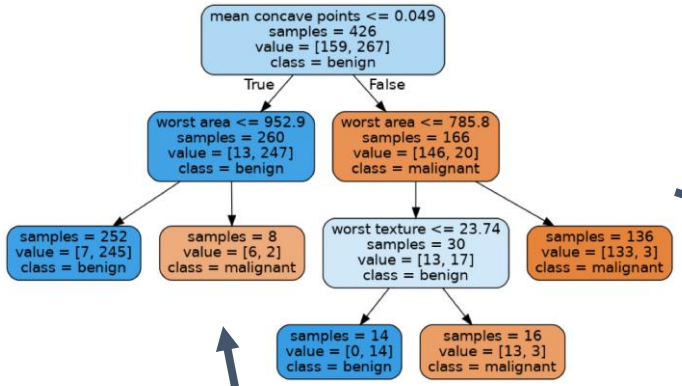


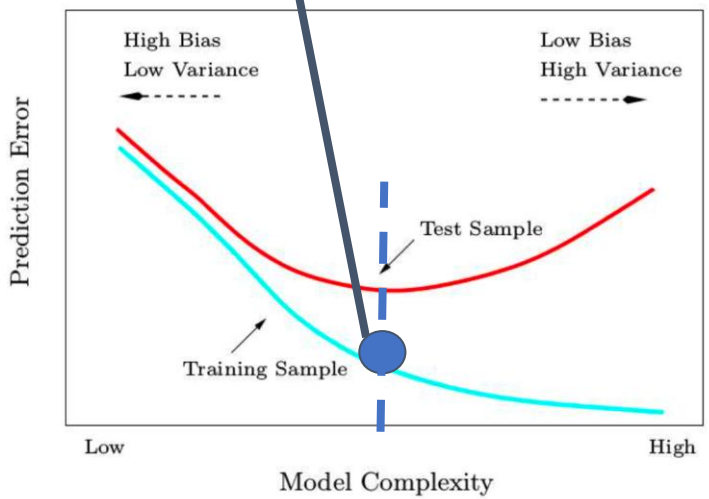


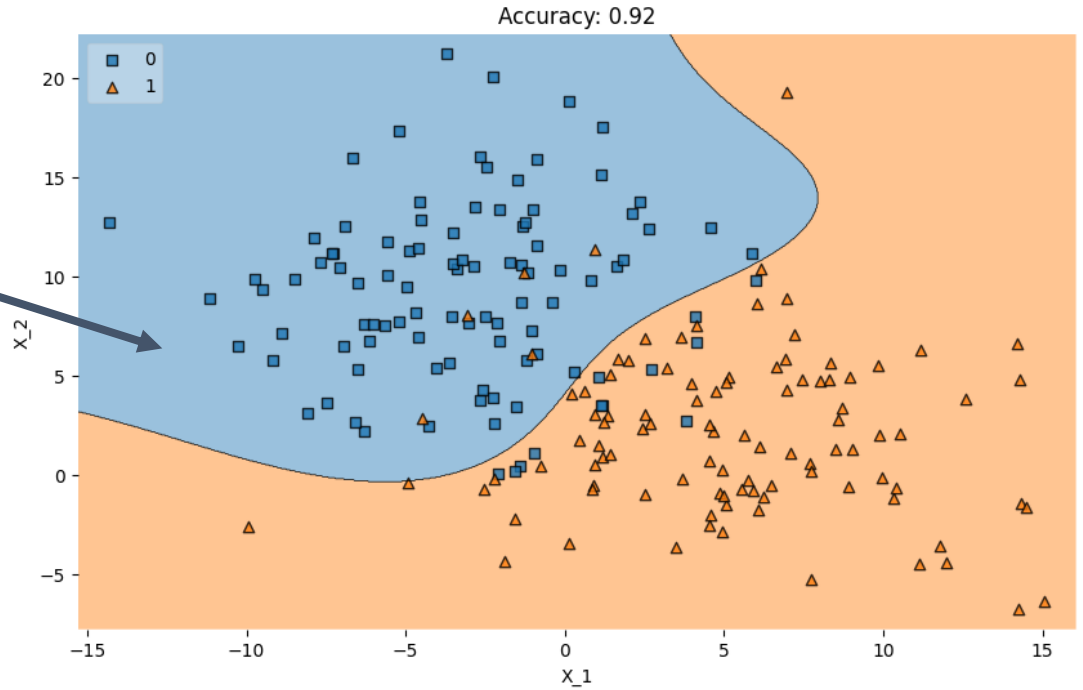
BiotrAIn

Bias-variance tradeoff

A good model

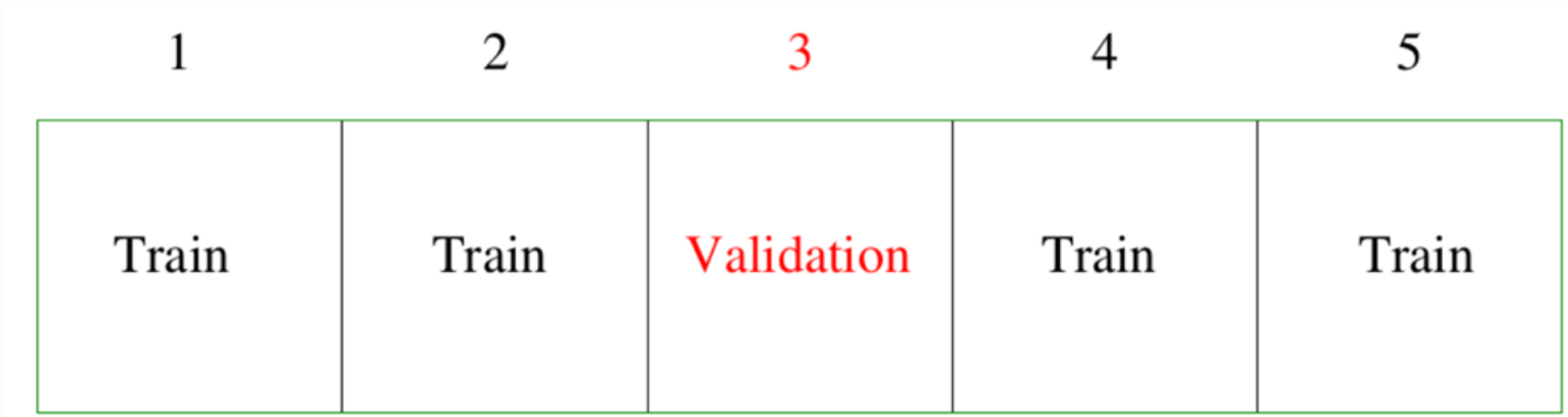






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Techniques such as **cross validation** (K-Fold CV, Stratified K-Fold CV, leave-one-out), are used to assess a model's performance by splitting the dataset into multiple subsets for training and testing. It helps evaluate how well a model generalizes to unseen data and reduces overfitting.



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Imbalanced data

An imbalanced dataset occurs when some classes contain significantly more samples than others. This can bias machine learning models toward the majority class, leading to poor performance on minority classes.

Options:

- **Class weighting:** Assign higher penalties to errors on minority classes.
- **Oversampling (data augmentation):** Increase minority-class samples.
- **Undersampling:** Reduce majority-class samples.
- **Threshold adjustment:** Modify the decision threshold to improve minority-class detection.
- **Appropriate metrics:** Use F1-score, Balanced Accuracy, ROC-AUC, PR-AUC, or Cohen's Kappa instead of accuracy alone.

BiotrAIn

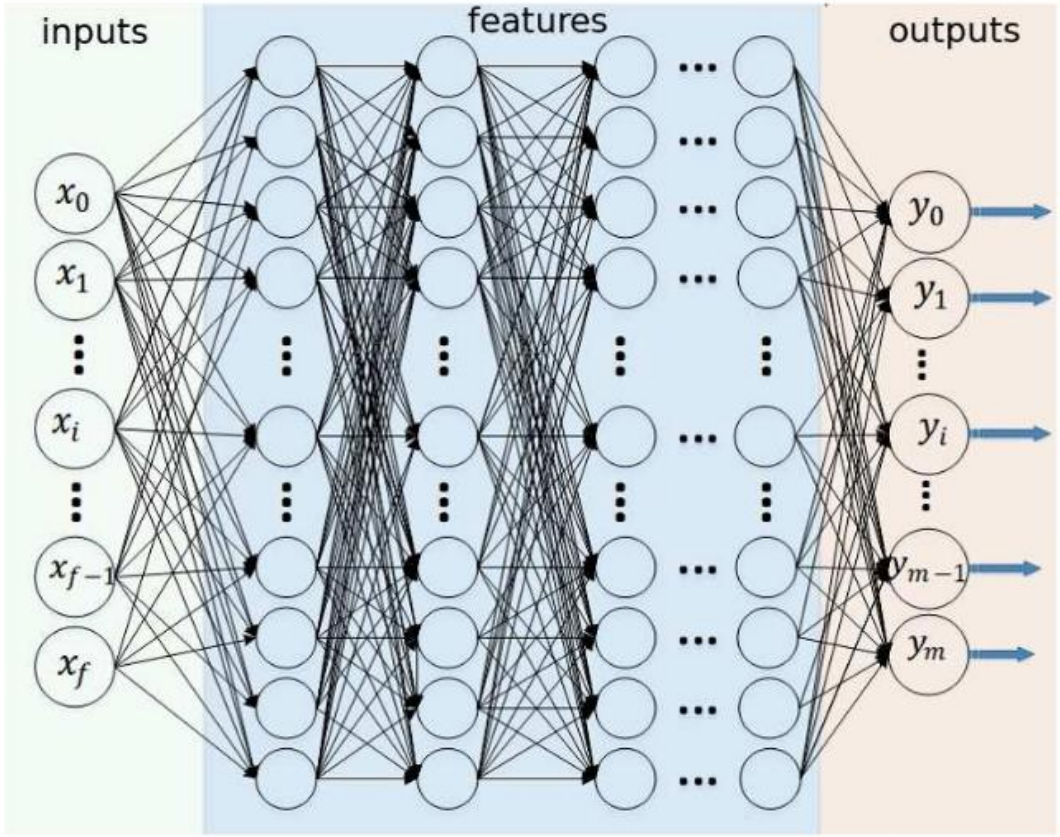
Imbalanced data

Example with SVM and Gaussian kernel



BiotrAIn

Deep learning models

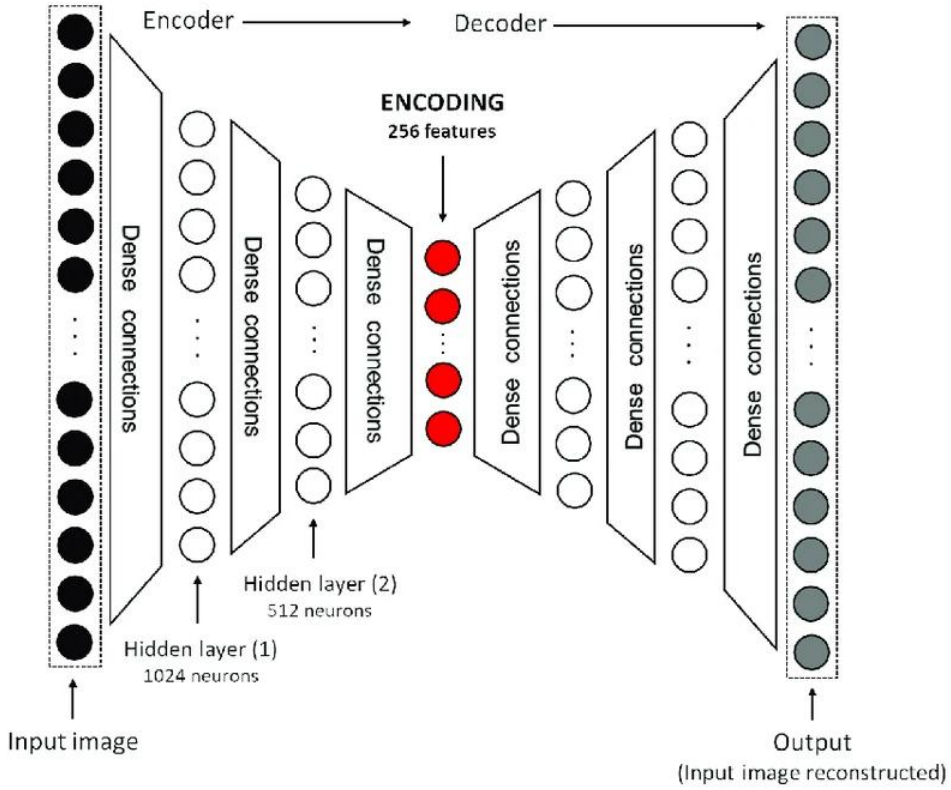


As the network gets deeper, it requires more data…

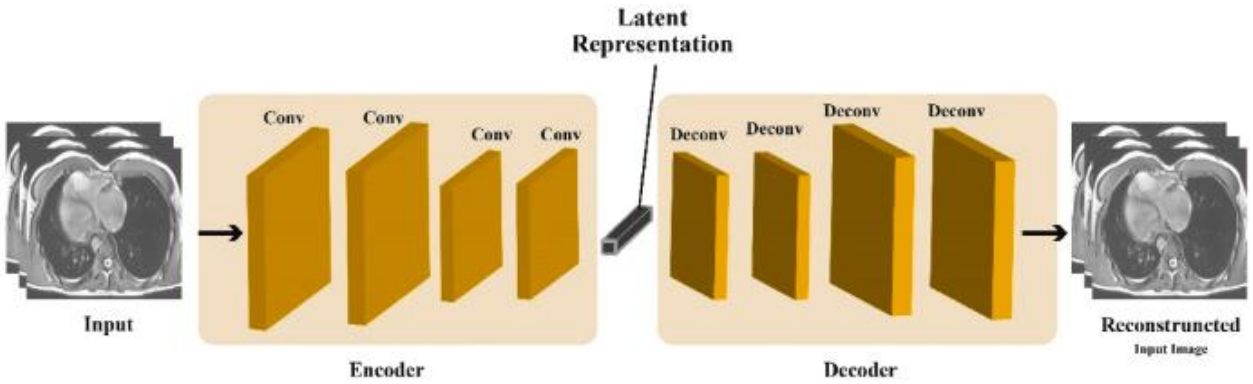
BiotrAIn

Deep autoencoder

A neural network that compresses input data into a low-dimensional latent representation (**encoder**) and then reconstructs the original data from that representation (**decoder**). The bottleneck layer forces the model to **learn the most important features of the data**.



Example: image reconstruction



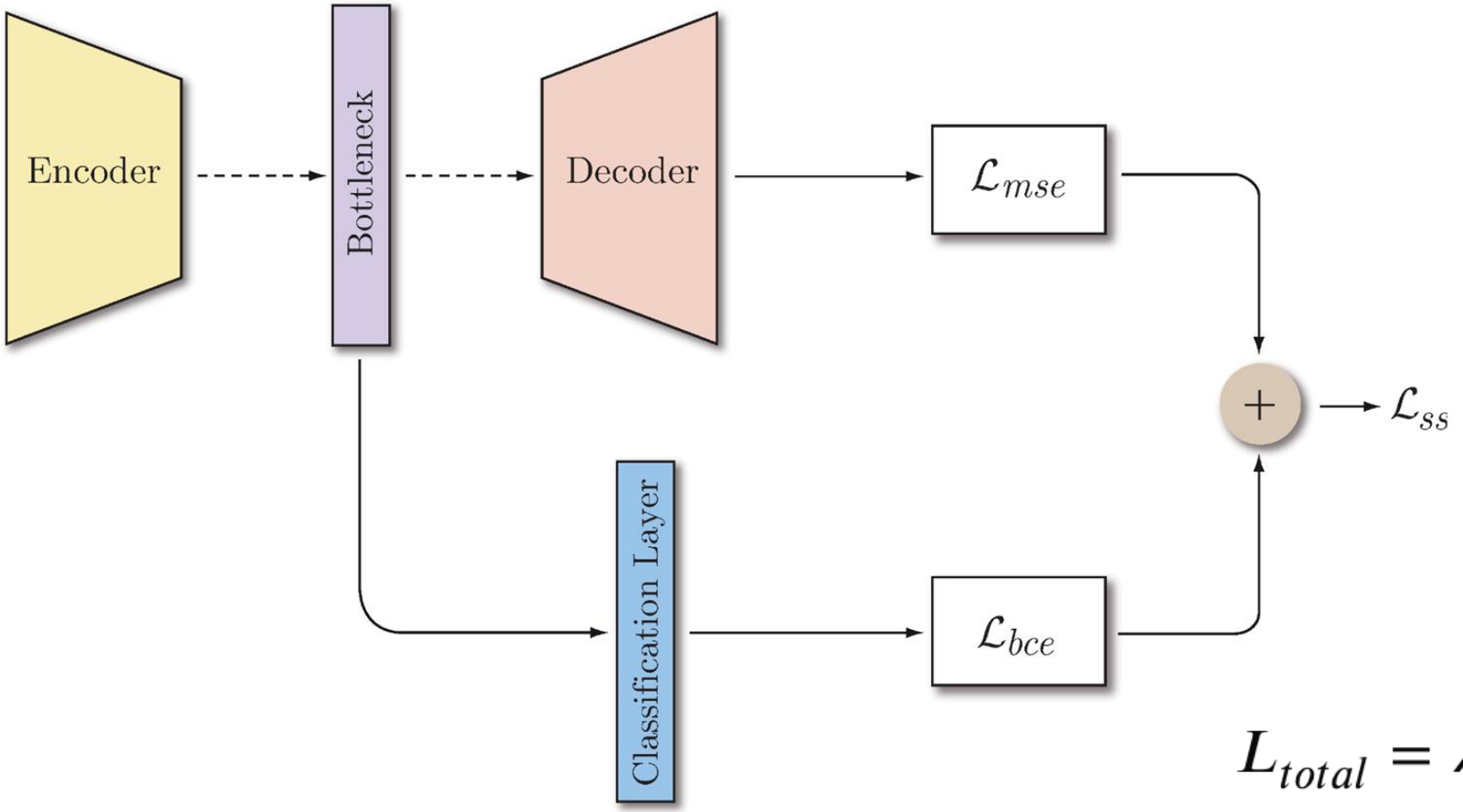
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Classifier Autoencoder (CAE) is a neural network that combines an **autoencoder** and a **classifier** in a single architecture. It learns a low-dimensional latent representation of the data while simultaneously optimizing classification performance.

In **gene expression data**, where the number of genes (features) is often much larger than the number of samples, a CAE can compress thousands of gene expression measurements into a compact latent space that preserves both the main biological variation and the information needed to distinguish classes (e.g., cancer types, disease states, or treatment responses).

BiotrAIn

Deep learning models



$$L_{total} = \lambda L_{recon} + (1 - \lambda) L_{class}$$

BiotrAIn